COMSATS University Islamabad,
Sahiwal Campus

NutriSafe

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By

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CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS (SE) “NutriSafe: An AI-Powered Food Inspection and Health Awareness System” was developed by **Rabeea Shahzad (CIIT/FA22-BSSE-124)** and **Areeba Akram** (**CIIT/FA22-BSSE -127)** under the supervision of “**Mr. Fahad Amin**” and co supervisor “**Miss Hafsa Saleem**” and that in (their/his/her) opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Computer Sciences.

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Executive Summary

The increasing use of packaged and processed foods has led to a rise in health issues such as diabetes, hypertension, allergies, and digestive disorders. Many consumers unknowingly consume harmful ingredients due to difficulty in understanding nutritional labels, while existing food scanning applications lack personalization and real-time health insights.

NutriSafe is an AI-powered mobile application designed to help users make safer dietary choices. It integrates Artificial Intelligence, Machine Learning, Optical Character Recognition (OCR), and Natural Language Processing (NLP) into a user-friendly Android platform. The system allows users to scan food packaging using a camera or barcode. OCR extracts ingredient data, which is then analyzed against the user's personalized health profile, including medical conditions and allergies. Based on this analysis, products are categorized as Safe, Caution, or Avoid, with clear explanations of harmful ingredients.

The application also suggests healthier alternatives for unsuitable products and enables users to report symptoms after consumption. Machine learning models use this feedback to predict future health risks and provide proactive warnings. An NLP-based chatbot offers real-time dietary guidance and improves user engagement through interactive support. The system continuously learns from user behavior and feedback, enhancing the accuracy of recommendations over time. This adaptive learning approach ensures that the system becomes more reliable and personalized with continued use.

NutriSafe is developed using Kotlin, Python, and TensorFlow, and operates in a secure cloud environment with AES-256 encryption and role-based access control. An admin panel manages the ingredient database, validates recommendations, and ensures system reliability. The platform is designed to be scalable, efficient, and accessible to a wide range of users. It can also be extended to integrate with wearable devices and health monitoring systems in the future.

Acknowledgement

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Rabeea Shahzad   Areeba Akram

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Abbreviations

SRS	Software Requirements Specification
SDD	Software Design Description
OCR	Optical Character Recognition
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
API	Application Programming Interface
UI	User Interface
HCI	Human-Computer Interaction
UCD	User-Centered Design
ERD	Entity Relationship Diagram
AES	Advanced Encryption Standard
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
FDA	Food and Drug Administration
NHS	National Health Service
PC	Personal Computer
DB	Database
FR	Functional Requirement
NFR	Non-Functional Requirement

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Chapter 1
Introduction

1. Introduction

This chapter gives an overview of the NutriSafe project. It explains what the project is about, why it was made, how it connects to university courses, what other similar systems exist, and what development approach was used to build it.

1.1 Brief

In today's world, people are becoming more and more concerned about what they eat. Many individuals suffer from health problems like diabetes, high blood pressure, heart disease, and food allergies. These conditions require people to be very careful about the food products they consume. However, reading and understanding food labels is not easy. Ingredient lists are often written in complex scientific language that most people cannot understand quickly or easily.

NutriSafe is an Android-based mobile application that solves this problem using Artificial Intelligence (AI). The main idea behind NutriSafe is simple a user takes a photo of a food product's packaging using their mobile phone, and the application automatically reads the ingredients, checks them against the user's personal health information, and tells the user whether that food is safe for them to eat or not.

The application works in the following way. When a user scans a food product image, the system uses a technology called Optical Character Recognition (OCR) to read and extract the list of ingredients from the image. These ingredients are then compared with the user's health profile, which contains information about their medical conditions, allergies, and dietary restrictions. Based on this comparison, the system labels the product as "Safe," "Caution," or "Avoid." If a product is not safe for the user, NutriSafe automatically suggests healthier and safer food alternatives.

NutriSafe also has a feedback feature. After eating a food item, users can report any symptoms or health reactions they experienced. The system saves this information and uses machine learning to predict future health risks if the user tries to consume similar products again. This makes NutriSafe not just a food scanner, but a smart health companion that learns from the user's experiences over time.

Additionally, NutriSafe includes an AI-powered chatbot. Users can type questions in plain English, such as "Is this ingredient safe for a diabetic person?" or "What foods should I avoid with high blood pressure?" and the chatbot will provide helpful, personalized answers instantly.

The application is built using Kotlin for the Android front-end, Python for artificial intelligence and machine learning, Tesseract OCR for reading text from images, and TensorFlow for building deep learning models. The project follows the Agile development methodology, which allows the team to build and improve the system step by step. NutriSafe is designed for people with chronic illnesses, food allergies, and anyone who wants to make healthier and more informed food choices in their daily life[1].

1.2 Relevance to Course Modules

The NutriSafe project is directly connected to many courses studied during the Bachelor of Science in Computer Science program. Below is an explanation of how each relevant course relates to this project:

Artificial Intelligence (AI): AI is at the heart of NutriSafe. The chatbot uses Natural Language Processing (NLP) to understand and respond to user questions. The recommendation engine uses AI logic to suggest safer food alternatives. The compatibility checker uses AI rules to evaluate ingredient safety based on the user's health profile [1].

Machine Learning (ML): The health risk prediction module is built using machine learning. The system analyzes the user's past feedback and symptom reports to identify patterns in which ingredients cause negative reactions. It then uses this knowledge to warn the user before they consume similar harmful products in the future. The ingredient detection module also uses image processing techniques to handle and process food packaging photos taken by the user. The OCR engine reads text from these images, which requires image enhancement, noise removal, and text recognition all core concepts of image processing.

Software Engineering: The entire NutriSafe project follows proper software engineering practices. This includes writing a Scope Document, Software Requirements Specification (SRS), designing use case diagrams, following the Agile development lifecycle, and preparing proper project documentation at every stage.

Database Systems: NutriSafe stores important data such as user health profiles, scanned ingredient lists, scan history, symptom reports, and food recommendations. All of this data is managed using a structured database system with encrypted storage to keep user information safe and private.

Mobile Application Development: The user-facing side of NutriSafe is a native Android application built using the Kotlin programming language. Concepts from mobile application development, such as activity management, camera integration, API communication, and UI design, are all applied in this project.

Human-Computer Interaction (HCI): The design of NutriSafe's user interface follows HCI principles. The goal is to make the application easy and comfortable to use for everyone, including people who are not technically skilled. Clear icons, simple menus, readable fonts, and helpful error messages are all part of making the application user-friendly [2].

Computer Networks: NutriSafe communicates with cloud-based AI servers and external food databases through the internet. Concepts from computer networks, such as HTTP communication, API requests, and data security during transmission, are applied throughout the system.

1.3 Project Background

The idea for NutriSafe came from a very real and common problem that millions of people face every day. When someone goes to a supermarket and picks up a food product, they often do not have the time, knowledge, or tools to understand what is written on the ingredient label. For a healthy person, this might not matter much. But for someone with diabetes, kidney disease, a nut allergy, or high blood pressure, eating the wrong ingredient can have serious health consequences.

According to health research, diet-related diseases are among the leading causes of illness and death worldwide. Conditions like diabetes, hypertension, heart disease, and obesity are largely influenced by what people eat. Despite this, most people do not have access to quick, personalized, and reliable tools that can help them make safe food choices.

Existing food scanning apps do exist, but they have major limitations. Most of them simply display the ingredient list without telling the user whether those ingredients are safe for their specific health condition. They do not offer personalized advice, they do not suggest alternatives, and they do not learn from the user's past experiences. Furthermore, most apps do not have any conversational support, leaving users confused when they have questions about a specific ingredient or food item.

NutriSafe was designed to fill all of these gaps. The project aims to create a complete, intelligent, and personalized food safety companion that combines the power of AI, machine learning, image processing, and natural language processing into one simple and easy-to-use mobile application. The system is not just about detecting harmful ingredients it is about empowering users to take control of their health through smarter, more informed food choices every single day.

1.4 Literature Review

Before building NutriSafe, a detailed study of existing food inspection and health-related applications was conducted. This helped the team understand what is already available in the market, what works well, and most importantly, what is missing. Below is a review of three major existing systems along with relevant research in the field.

Food Check: AI Product Scanner Food Check is another food scanning application that uses some AI features to analyze food products. It retrieves ingredient information and provides a general safety rating. While it is more advanced than basic scanner apps, it still has significant gaps. It does not cross-reference ingredients with a user's personal health profile in real time. It lacks an NLP-based user feedback system, meaning it cannot learn from what users report after consuming a product. It also does not recommend specific alternative food products when a scanned item is found to be unsafe, and it provides no conversational user support.

Ingredio Ingredients Scanner Ingredio is an application focused on scanning food product labels and displaying ingredient information. It provides some basic warnings about certain additives but does not support personalized health profiles. It has no mechanism for users to report health reactions after eating. It does not offer any food analysis forecasting or predictive health risk features, and it provides no chatbot or user support system.

Research and Academic Background Beyond existing apps, academic research supports the approach taken by NutriSafe. Studies in the field of food informatics have shown that OCR-based ingredient extraction from food packaging is highly feasible using modern deep learning models, with recognition accuracy improving significantly when preprocessing techniques are applied to enhance image quality. Research in NLP-based health chatbots has demonstrated that conversational agents can effectively improve user engagement and support better dietary decision-making, particularly for people managing chronic conditions. Machine learning models trained on user-reported symptom data have also been successfully used in predictive health systems to identify patterns between food consumption and health outcomes, further validating the approach used in NutriSafe's health risk prediction module.

1.5 Analysis from Literature Review (in the context of NutriSafe)

After reviewing existing systems and related research, it is clear that while several food scanning tools exist, none of them offers a fully integrated and personalized solution. The following key gaps were identified:

No Personalization: Existing apps provide the same information to all users. They do not take individual health conditions, allergies, or dietary restrictions into account when evaluating a food product.

No Feedback Loop: None of the reviewed apps allow users to report symptoms after consuming a product. This means the apps cannot learn from user experiences or improve their predictions over time.

No Alternative Recommendations: When a product is found to be unsafe, existing apps simply warn the user but do not suggest what they should eat instead. This leaves users without a practical next step.

No Real-Time Support: Existing apps lack conversational chatbot features, making it difficult for users to get instant answers to their food-related questions.

No Predictive Health Features: Current systems are entirely reactive they only respond to what is scanned at that moment. They do not use past data to predict future health risks. NutriSafe directly addresses every one of these gaps. It provides fully personalized health compatibility analysis, a feedback-driven learning loop, AI-powered food alternatives, an NLP chatbot for real-time support, and machine learning-based health risk prediction. In this way, NutriSafe goes far beyond any existing system and represents a genuinely new and valuable contribution to the field of personal health and food safety technology.

1.6 Methodology and Software Lifecycle for This Project

Every software project needs a clear plan and a structured way of working. This plan is called a Software Development Methodology, and it defines how the project will be organized, how tasks will be divided, and how progress will be tracked from start to finish. For NutriSafe, the team selected the Agile Software Development Methodology as the primary development approach. Agile is a modern and flexible way of building software that divides the entire project into small, manageable phases called "sprints." Each sprint focuses on developing and testing a specific part of the system. After each sprint, the team reviews the work done, collects feedback, and makes improvements before moving on to the next phase.

In addition to Agile, the team also applied User-Centered Design (UCD) principles throughout the development process. This means that at every stage of the project, the needs, preferences, and experiences of the end users were kept at the center of all design and development decisions. Regular usability checks were performed to make sure the application remained easy to understand and comfortable to use.

The overall software lifecycle for NutriSafe followed these main phases:

Phase 1: Requirement Gathering and Analysis: The team identified the needs of the target users, studied existing systems, and documented all functional and non-functional requirements in the Scope Document and SRS.

Phase 2: System Design: The team designed the overall system architecture, created use case diagrams, planned the database structure, and designed the user interface mockups using Figma [3].

Phase 3: Iterative Implementation: Each module of NutriSafe including ingredient detection, health profile management, compatibility checking, recommendation engine, symptom feedback, health risk prediction, and the chatbot was developed one by one in separate sprints.

Phase 4: Testing and Evaluation: Each module was individually tested after development (unit testing), then tested as part of the complete system (integration testing), and finally evaluated against all documented requirements (functional testing).

Phase 5: Deployment and Finalization: The completed application was packaged and prepared for deployment on the Android platform through the Google Play Store.

1.6.1 Rationale behind Selected Methodology

The Agile methodology was chosen for NutriSafe for the following reasons:

Flexibility: NutriSafe involves several AI and machine learning components. The performance of these components depends heavily on the quality of training data and model tuning, which cannot always be fully predicted in advance. Agile's flexible structure allows the team to adjust requirements and designs as new findings emerge during development without disrupting the entire project.

Iterative Development: Because NutriSafe is made up of multiple independent modules, Agile's sprint-based approach allows each module to be built and tested separately. This reduces the risk of large-scale integration failures and makes it easier to identify and fix problems early.

Continuous Feedback: Agile includes regular review meetings and feedback sessions with the project supervisor. This ensures that the system stays aligned with academic and technical requirements throughout the entire development process.

User Focus: By combining Agile with User-Centered Design, the team was able to continuously check that the application remains easy to use and meets the real needs of its target users. This is especially important for NutriSafe, which is designed for everyday users who may not have any technical background.

Suitability for Small Teams: Agile is particularly well-suited to small development teams like the NutriSafe team, which consists of two members. Its lightweight planning and documentation approach allows the team to stay productive and focused without being overwhelmed by heavy administrative processes.

Chapter 2
Problem Definition

2. Problem Definition

This chapter explains the exact problem that NutriSafe is designed to solve. It describes the current situation, why existing solutions are not enough, what NutriSafe will deliver, and what the current systems look like before NutriSafe was introduced.

2.1 Problem Statement

Food is one of the most basic and essential needs of every human being. Every day, people go to stores, pick up food products, and consume them without fully understanding what those products contain. For a completely healthy person, this may not always cause immediate harm. However, for the millions of people around the world who suffer from chronic health conditions such as diabetes, high blood pressure, heart disease, kidney problems, or food allergies, consuming the wrong ingredient even in a small amount can lead to serious and sometimes life-threatening health consequences.

The core problem is that most people simply do not have the knowledge, time, or tools to properly evaluate the safety of a food product before eating it. Food labels are often written in complex scientific language filled with chemical names, preservative codes, and additive numbers that are very difficult for ordinary people to understand. Even educated individuals struggle to quickly determine whether a particular ingredient is harmful to their specific health condition while standing in a supermarket aisle.

This problem is made worse by several factors that exist in today's food environment:

Lack of Awareness: Many people are completely unaware of which ingredients are harmful to their particular health condition. For example, a person with kidney disease may not know that certain preservatives can worsen their condition, or a person with diabetes may not realize that a product labeled as "healthy" still contains hidden sugars under different chemical names.

Complex Food Labels: Food packaging often contains dozens of ingredients listed in very small text using technical scientific terminology. It is practically impossible for most consumers to read, understand, and evaluate all of these ingredients quickly and accurately without professional help.

No Personalized Tools: Most food scanning apps available today provide general information about a product but do not personalize their analysis based on the individual user's health profile. They tell everyone the same thing regardless of whether that person has diabetes, a gluten allergy, or high blood pressure. This generic approach is not useful for people with specific health conditions who need tailored guidance.

No Real-Time Guidance: When a person is unsure about a food ingredient or product, they have no quick and reliable source of guidance available to them. They may search the internet, but this takes time and often returns confusing or contradictory results. No existing app provides instant, personalized, conversational support through a chatbot that can answer specific dietary questions in real time.

No Symptom Tracking or Risk Prediction: People who experience negative health reactions after eating certain foods often have no way of recording those reactions and linking them to specific ingredients. As a result, they may unknowingly consume the same harmful ingredient again and again, each time suffering the same negative effects without understanding the cause. No existing system learns from a user's past experiences and proactively warns them about future risks.

Difficulty for Vulnerable Groups: People with low health literacy, elderly individuals, and those managing multiple health conditions at the same time face the greatest difficulty in making safe food choices. These vulnerable groups need the most support but currently have the least access to personalized, easy-to-use food safety tools.

In summary, the core problem that NutriSafe addresses is the absence of a smart, personalized, real-time, and AI-powered food safety system that can help individuals especially those with chronic health conditions make informed, safe, and healthy food choices every single day based on their unique health needs.

2.2 Deliverables and Development Requirements

This section outlines the key deliverables of the NutriSafe project along with the technical and development requirements needed for successful implementation. It includes the mobile application, backend services, documentation, and necessary tools, technologies, and resources required to design, develop, and deploy the system efficiently.

2.2.1 Deliverables

The NutriSafe project will deliver the following outputs upon completion:

1. A Fully Functional Android Mobile Application The primary deliverable of this project is a working Android application called NutriSafe. The application will be installable on any Android device running Android version 10.0 or higher. It will include all seven core

modules described in the project scope and will be ready for real-world use by the target audience.

2. Ingredient Detection Module This module will allow users to take a photo of any food product's packaging using their mobile phone camera. The system will use Optical Character Recognition (OCR) technology powered by Tesseract OCR to automatically read and extract the list of ingredients from the image. The extracted ingredients will be cleaned, filtered, and prepared for further analysis.

3. Personalized Health Profile Module Users will be able to create and manage their own health profiles within the application. This profile will store important personal health information including known medical conditions such as diabetes or hypertension, food allergies such as nut or gluten allergy, and specific dietary restrictions. This profile forms the foundation for all personalized analysis performed by the system.

4. Food Compatibility Checker Module Once ingredients are extracted from a food product, this module will compare them against the user's health profile. Each ingredient will be evaluated and the product will be labeled as one of three categories: "Safe" meaning the product is suitable for the user, "Caution" meaning the product contains ingredients that may cause mild concern, or "Avoid" meaning the product contains ingredients that are clearly harmful for the user's specific health condition.

5. AI-Powered Food Recommendation Module When a product is labeled as "Caution" or "Avoid," this module will automatically suggest healthier and safer alternative food products. These recommendations will be sourced from verified nutritional databases and will be tailored to match the user's health profile and food preferences as closely as possible.

6. Symptom Feedback Module After consuming a food item, users will be able to open the application and report any symptoms or health reactions they experienced. They will select the food item from their scan history and describe their symptoms through a simple feedback form. This information will be stored in the system and linked to specific ingredients in that product, creating a personal database of the user's food reactions over time.

7. Health Risk Prediction Module Using the symptom feedback data collected over time, this module will apply machine learning models to identify patterns between specific ingredients and negative health reactions. When the system detects that a new food product contains an ingredient that previously caused the user a health problem, it will proactively warn the user before they consume it, effectively turning NutriSafe into a preventive health tool rather than just a reactive one.

8. AI Chatbot Module An NLP-based conversational chatbot will be integrated into the application. Users will be able to type questions in plain, everyday English and receive instant, helpful, and personalized answers about food ingredients, dietary advice, and food safety. For example, a user can ask "Can I eat this with high blood pressure?" or "What does this ingredient do to the kidneys?" and the chatbot will respond clearly and helpfully.

9. Project Documentation All required academic and technical documents will be delivered as part of this project, including the Scope Document, Software Requirements Specification (SRS), Software Design Description (SDD), Project Report, and Testing Documentation.

2.2.2 Development Requirement

To successfully develop the NutriSafe application, the following development requirements must be met:

Hardware Requirements:

• A development computer with sufficient processing power to run Android Studio, Python environments, and machine learning model training tasks.

• An Android smartphone or emulator with Android version 10.0 or higher for testing the application during development.

• A functional rear camera on the test device for testing the OCR-based ingredient scanning feature.

Software Requirements:

• Android Studio with Kotlin support for developing the Android mobile application [4].

• Python 3.x for developing AI, machine learning, and NLP modules.

• Tesseract OCR version 5.5.1 for text extraction from food packaging images.

• TensorFlow version 2.19.0 for building and training machine learning models used in health risk prediction and ingredient analysis [5].

• Figma for designing user interface prototypes and mockups.

• Git and GitHub for version control and collaborative code management throughout the development process.

Data Requirements:

• Open Food Facts API to retrieve detailed ingredient and nutritional information for food products.

• FDA and NHS ingredient warning databases to identify and flag harmful or restricted ingredients.

• User-reported symptom data collected through the feedback module, used for training the health risk prediction machine learning model.

• Recipe and food alternative APIs to support the recommendation module in suggesting safer food options.

Network Requirements:

• A stable internet connection is required for the application to communicate with cloud-based AI servers, the OCR processing engine, the Open Food Facts API, and the NLP chatbot model [6].

• All data transmitted between the application and external servers must be secured using HTTPS protocols to protect user privacy.

2.3 Current System (if applicable to your project)

Before NutriSafe was developed, there was no single, integrated solution available that could address all the food safety challenges described in the problem statement. However, several existing applications were studied to understand what the current landscape looks like and where the biggest gaps are. The following is a description of the current systems that exist in this space:

2.3.1 Existing Health Food Scanner Applications

Several health food scanner applications are currently available on mobile platforms that allow users to scan food barcodes and retrieve basic nutritional information. These applications typically display general data such as calorie counts, fat content, sugar levels, and an overall health score to help users understand the nutritional value of a product.

However, these systems have significant limitations. They do not incorporate personalized user health profiles, meaning that all users receive the same information regardless of their individual medical conditions, allergies, or dietary restrictions. Furthermore, such applications lack advanced features such as food compatibility analysis based on health conditions, intelligent alternative product recommendations, symptom reporting mechanisms, and interactive chatbot support.

In addition, these systems are largely static in nature, as they do not utilize machine learning or adaptive techniques to improve recommendations based on user behavior or health history. As a result, their ability to provide meaningful and personalized health guidance is limited.

2.3.2 Food Check: AI Product Scanner

Food Check is another currently available application that uses some AI capabilities to scan food products and provide ingredient analysis. It retrieves ingredient data and gives a general rating for the product. While it is slightly more advanced , it still lacks personalized rather than the specific health needs of the individual user. It has no NLP-based feedback system, no predictive health risk features, and no mechanism for recommending alternative products when a scanned item is found unsuitable. Real-time chatbot support is also completely absent.

2.3.3 Ingredio Ingredients Scanner

Ingredio focuses on scanning food product labels and displaying the list of detected ingredients along with basic information about certain additives. It is primarily an information display tool rather than a true health analysis system. The current version of Ingredio does not support personalized health profiles, has no user support or chatbot feature, offers no predictions or forecasting about future health risks, and cannot recommend alternative food products. It is a passive system that simply shows data without providing any personalized or actionable health guidance.

2.3.4 Summary of Current System Limitations

The table below summarizes the key features that are missing from all current systems and how NutriSafe addresses each one:

As clearly shown in the table above, NutriSafe is the only system that addresses all of these critical needs in one integrated application. It does not simply improve upon existing systems it introduces an entirely new level of personalized, predictive, and preventive food safety intelligence that no current system provides.

Chapter 3
Requirement Analysis

3. Requirement Analysis

This chapter presents the complete requirement analysis for the NutriSafe system. It begins with an introduction to the requirement analysis process and explains why the Use Case technique was selected for NutriSafe. It then describes the system overview and all actors involved, presents the use case diagram, and provides structured and detailed descriptions of every major use case. The chapter concludes with a comprehensive listing of the system's functional and non-functional requirements. This chapter serves as the definitive technical reference for what NutriSafe must do and the standards it must meet.

3.1 Introduction to Requirement Analysis

Requirement analysis is one of the most critical phases in the software development lifecycle. It is the process through which the development team identifies, documents, and validates the needs and expectations of all users and stakeholders of a software system before any implementation begins. When requirement analysis is performed thoroughly and correctly, it ensures that the final software product solves the right problems, delivers the right features, and meets the expectations of everyone who will use or depend on it.

For a system as complex and multi-layered as NutriSafe, which involves artificial intelligence, image processing, machine learning, and natural language processing components, careful requirement analysis is especially important. The team conducted requirement analysis through a combination of studying existing food inspection applications, reviewing related academic research, consulting with the project supervisor, and evaluating user needs through targeted research on the health challenges faced by people with chronic illnesses and food allergies.

Several requirement identification techniques exist in software engineering, each suited to different types of systems. Use Case Diagrams are most appropriate for interactive systems where users perform specific actions and expect defined system responses. Event-Response Tables are better suited to real-time systems that react to environmental inputs. Storyboarding is useful for visually rich applications such as games or media tools.

For NutriSafe, the Use Case technique was selected as the primary requirement identification method. This technique is ideal for NutriSafe because the system is fundamentally interaction-driven: users scan food products, check compatibility, receive recommendations, chat with an

AI assistant, and submit feedback. Use cases capture all of these interactions in a clear, structured, and easily understandable format that can be reviewed and validated by both technical and non-technical stakeholders. The use case diagrams and their detailed descriptions together provide a complete and unambiguous picture of every major function the system must perform.

3.2 System Overview

NutriSafe is an Android-based intelligent food inspection and health guidance system that serves individuals who need personalized dietary analysis based on their specific health conditions. At a high level, the system allows users to photograph food product packaging, automatically extract and analyze the ingredient list, and receive a personalized safety assessment based on their stored health profile. When a product is found to be incompatible, the system recommends safer alternatives. Users can also report symptoms after consumption, and the system uses this feedback to predict and proactively warn about future health risks. An AI-powered chatbot provides real-time dietary support throughout the user's experience.

The system is divided into seven core functional modules: Ingredient Detection, Health Profile Management, Food Compatibility Checking, AI-Powered Recommendation, Symptom Feedback, Health Risk Prediction, and AI Chatbot. These modules are supported by an Administrator backend panel that manages the ingredient database, validates AI-generated recommendations, and monitors overall system performance. All data is stored securely using AES-256 encryption, and all external communications use HTTPS protocols to protect sensitive user health information.

3.3 Actors in the System

The NutriSafe system involves three distinct actors whose interactions define the boundaries and behaviors of the application. Each actor has a specific role and a clearly defined set of responsibilities within the system.

3.3.1 User

The User is the primary actor in the NutriSafe system. This role represents any individual who downloads and uses the NutriSafe application on their Android smartphone. Users are typically people with chronic health conditions such as diabetes, hypertension, or kidney disease, individuals with food allergies, or health-conscious people who want to better understand the ingredients in the products they consume. The User interacts with all core features of the application including registration, health profile setup, food scanning, compatibility results, food recommendations, chatbot queries, and symptom feedback submission. All personalized features of the system are built around the User's stored health data.

3.3.2 Admin

The Admin is a secondary actor who manages the operational backend of the NutriSafe system. This role is held by an authorized system administrator who is responsible for maintaining the accuracy and completeness of the ingredient and product database, reviewing and approving AI-generated food recommendations before they become visible to users, monitoring overall system performance including API response times and OCR accuracy rates, and managing user accounts. Admin access is restricted through role-based authentication and requires two-factor verification to prevent unauthorized access to sensitive system management functions.

3.3.3 External Service

The External Service actor represents the external APIs and databases that NutriSafe integrates with to retrieve food ingredient and nutritional information. The primary external service used by NutriSafe is the Open Food Facts API, which provides detailed ingredient lists and nutritional data for millions of food products. External services supplement the system's internal ingredient database and provide additional product information during the scanning and compatibility analysis processes. This actor is not a human but an automated external system that the NutriSafe Application Layer communicates with through secure API calls.

3.4 Use Case Diagram(s)

The Use Case Diagram of NutriSafe provides a high-level view of how different actors interact with the system. It visually represents the functionalities of the system and the relationships between users and system processes. The diagram plays an important role in understanding system behavior and identifying user interactions clearly.

The NutriSafe system consists of four primary actors: the User, the Admin, the Health Professional, and the External API. The User interacts with the system to perform core activities such as scanning food, checking compatibility, receiving recommendations, and interacting with the chatbot. The Admin is responsible for managing the ingredient database, validating system outputs, and monitoring overall performance. The Health Professional provides expert validation and ensures that recommendations are medically accurate. The External API supplies ingredient and product-related data, which enhances system accuracy and reliability.

The Use Case Diagram shows the interaction of these actors with key system functions including login, profile management, food scanning, compatibility analysis, recommendation generation, chatbot interaction, and feedback submission. This representation ensures a clear understanding of system operations and user involvement.

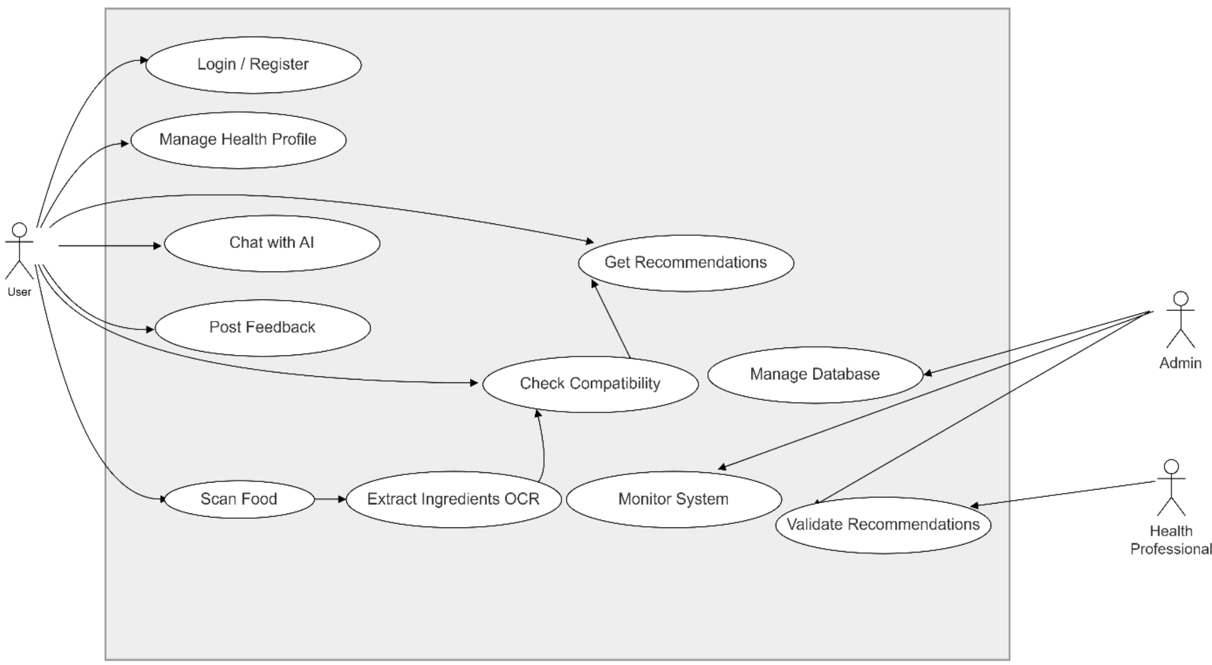

Figure 3.1:Use case diagram

3.5 Detailed Use Case Description

The detailed use case description provides a structured explanation of how each function of the system operates. It defines the interaction between the user and the system in a step-by-step manner, including preconditions, normal flow, alternative flow, and exceptions. This section helps in understanding the exact working of each feature and ensures that all functional requirements are clearly defined and documented.

Each use case is presented in a tabular format to maintain clarity and consistency. The following tables describe the major use cases of the NutriSafe system.

Table 3.1:Login/Register

Field	**Description**
Use Case ID	UC-01
Use Case Name	Login / Register
Actor	User
Description	The user registers or logs into the system using valid credentials to access personalized features
Pre-condition	The application must be installed and internet connection must be available
Post-condition	User is successfully logged in and dashboard is displayed
Normal Flow	The user opens the application, enters login credentials or registers a new account, and the system verifies the details before displaying the dashboard
Alternative Flow	If login fails, the system prompts the user to reset the password
Exceptions	Incorrect credentials or network failure

Table 3.2:Scan Food Ingredients

Field	**Description**
Use Case ID	UC-02
Use Case Name	Scan Food Ingredients
Actor	User, External API

Description	The user scans food packaging using the mobile camera and the system extracts ingredient data
Pre-condition	User must be logged in, and camera and internet must be active
Post-condition	Ingredient information is extracted and displayed
Normal Flow	The user captures an image, the system performs OCR, retrieves data from API, and displays the extracted ingredients
Alternative Flow	If OCR fails, the system allows manual input
Exceptions	Camera access denied or API not responding

Table 3.3:Check Food Compatibility

Field	**Description**
Use Case ID	UC-03
Use Case Name	Check Food Compatibility
Actor	User
Description	The system compares extracted ingredients with the user's health profile to determine compatibility
Pre-condition	Health profile must exist
Post-condition	Compatibility result is generated
Normal Flow	The system retrieves ingredient data, compares it with the health profile, and displays the result
Alternative Flow	The system requests additional user information if the profile is incomplete
Exceptions	Server or API failure

Table 3.4:Get Recommended Alternatives

Field	**Description**
Use Case ID	UC-04
Use Case Name	Get Recommended Alternatives
Actor	User
Description	The system suggests healthier alternatives when a food item is incompatible
Pre-condition	Compatibility check must be completed
Post-condition	Alternative food options are displayed
Normal Flow	The system identifies unsafe ingredients, retrieves alternatives, and displays them
Alternative Flow	The user can manually search for alternatives
Exceptions	No alternative data available

Table 3.5:Chat with AI

Field	**Description**
Use Case ID	UC-05
Use Case Name	Chat with AI
Actor	User
Description	The user interacts with an AI chatbot to ask questions related to food and health
Pre-condition	Internet connection must be active
Post-condition	AI-generated response is displayed
Normal Flow	The user enters a query, the system processes it, and generates a response
Alternative Flow	The chatbot suggests related queries

Exceptions	AI timeout or server failure

Table 3.6:Post Report

Field	**Description**
Use Case ID	UC-06
Use Case Name	Post Report
Actor	User
Description	The user reports symptoms or feedback after consuming a product
Pre-condition	User must be logged in
Post-condition	Report is stored in the system
Normal Flow	The user fills out the report form and submits it, after which the system stores the report
Alternative Flow	The user may attach additional notes or images
Exceptions	Network or server failure

3.6 Functional Requirements

Functional requirements define the core operations that the NutriSafe system must perform to meet user needs. These requirements describe what the system should do and how it should respond to user inputs. They are derived from the use cases and ensure that all system functionalities are clearly specified and implemented.

The functional requirements of NutriSafe are categorized based on different system modules including the User module, Admin module, and Health Professional module. These requirements collectively define the complete working behavior of the system.

The following table summarizes the functional requirements of the NutriSafe system.

Table 3.7:Functional Requirements

ID	Requirement Description
FR-01	The system shall allow users to register and log in
FR-02	The system shall allow users to create and update health profiles
FR-03	The system shall allow users to scan food using the camera
FR-04	The system shall extract ingredient information using OCR
FR-05	The system shall compare ingredients with the user's health profile
FR-06	The system shall classify food as Safe, Caution, or Avoid
FR-07	The system shall provide alternative food recommendations
FR-08	The system shall allow interaction with an AI chatbot
FR-09	The system shall allow users to submit feedback reports
FR-10	The system shall allow admin to manage ingredient database
FR-11	The system shall allow admin to validate recommendations
FR-12	The system shall monitor system performance
FR-13	The system shall allow health professionals to review recommendations
FR-14	The system shall ensure accurate health-based suggestions

3.7 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system, such as performance, security, usability, and reliability. These requirements do not describe specific functions but ensure that the system operates efficiently and meets user expectations in terms of quality and performance.

The NutriSafe system is designed to provide a user-friendly interface, fast processing speed, secure data handling, and reliable performance. These characteristics are essential for ensuring a smooth and effective user experience.

Table 3.8:Non-Functional Requirements

Category	Description
Usability	The system shall provide a simple and user-friendly interface that is easy to understand and use
Performance	The system shall respond quickly and process data in real time with minimal delay
Security	The system shall ensure data encryption and secure authentication to protect user information
Reliability	The system shall be available at all times with minimal downtime
Maintainability	The system shall be designed in a modular way to allow easy updates and maintenance
Scalability	The system shall support an increasing number of users and data efficiently

Chapter 4
Design and Architecture

4. Design and Architecture

The design and architecture of the NutriSafe system define the structural organization and operational flow of the application. This chapter explains how different components of the system interact with each other to deliver the required functionality. A well-designed architecture ensures scalability, maintainability, efficiency, and secure handling of user data. NutriSafe is designed using a modular and layered approach so that each component performs a specific function while remaining loosely coupled with other modules.

4.1 System Architecture

The system architecture of NutriSafe defines the overall structure of the application and how different components interact with each other. It provides a high-level view of the system design, showing how data flows between the user interface, processing units, and storage systems. The architecture ensures scalability, maintainability, and efficient communication between different modules.

NutriSafe follows a three-tier architecture, which separates the system into three main layers: the Presentation Layer, the Application Layer, and the Data Layer. This layered approach helps in organizing the system efficiently and allows independent development and maintenance of each component.

The Presentation Layer consists of the mobile application interface through which users interact with the system. It includes features such as login, profile management, food scanning, chatbot interaction, and viewing recommendations. This layer ensures a user-friendly experience and handles all user inputs and outputs.

The Application Layer contains the core logic of the system. It processes user inputs, performs OCR for ingredient extraction, analyzes compatibility using machine learning models, and generates recommendations. This layer acts as the brain of the system and connects the user interface with the database and external APIs.

The Data Layer is responsible for storing and managing all system data. It includes user profiles, ingredient databases, feedback reports, and recommendation data. This layer is implemented using cloud-based storage to ensure data availability, security, and scalability.

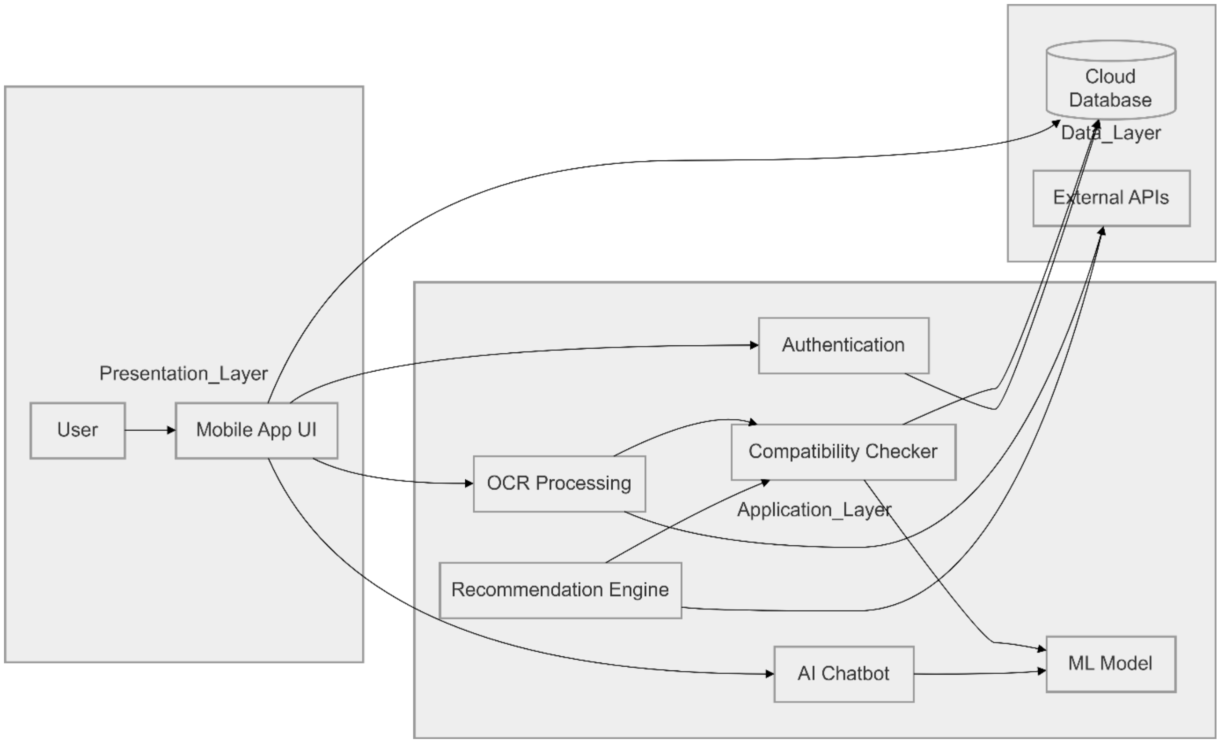

Figure 4.1:System Architecture of NutriSafe

4.2 Data Representation

Data representation describes how information is structured, stored, and managed within the system. In NutriSafe, data is organized in a way that supports efficient processing, retrieval, and analysis. The system uses structured and semi-structured data formats to handle different types of information such as user profiles, food ingredients, and health reports.

The main entities in the system include User, Health Profile, Food Item, Ingredient, Recommendation, and Feedback Report. Each entity stores specific information that is used in the processing and analysis of food compatibility.

The User entity contains personal information such as name, email, and login credentials. The Health Profile entity stores user-specific data such as allergies, diseases, and dietary restrictions. The Food Item entity includes details of scanned products, while the Ingredient entity stores extracted ingredient data. The Recommendation entity provides alternative food suggestions, and the Feedback entity stores user-reported symptoms.

These entities are interconnected to ensure that the system can analyze food items effectively and provide accurate recommendations. The relationships between these entities can be represented using an Entity-Relationship (ER) Diagram.

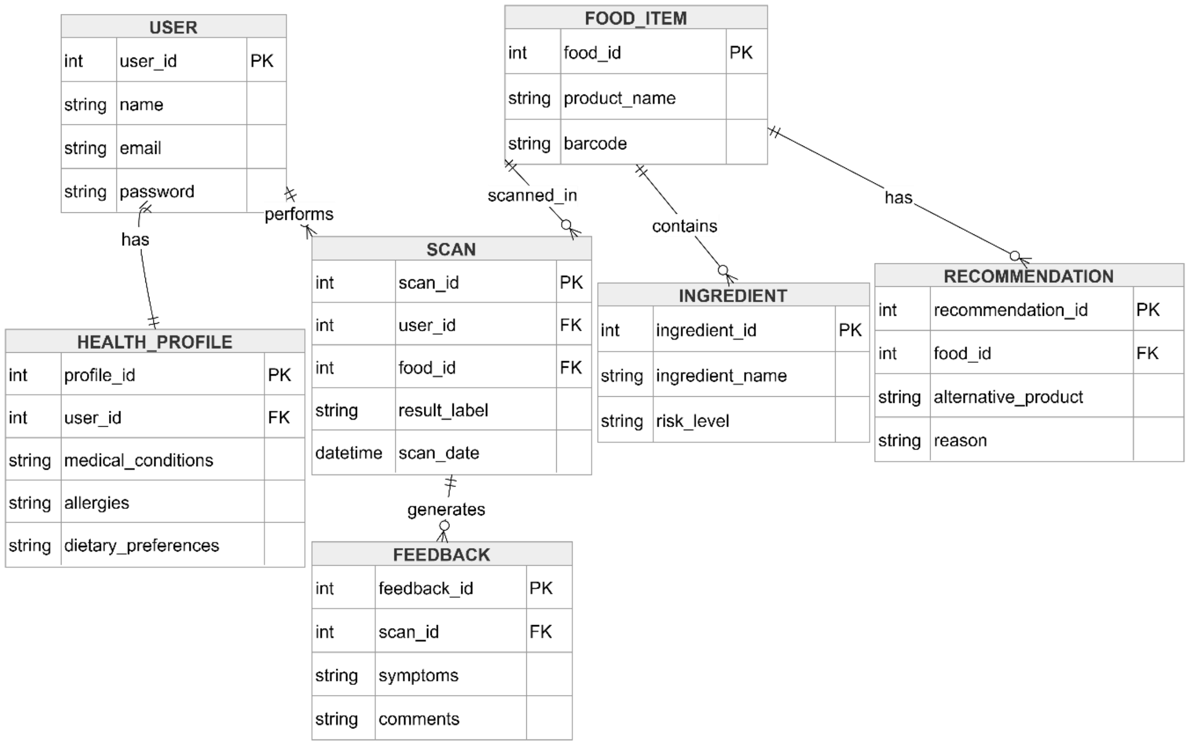

Figure 4.2:Entity Relationship Diagram (ERD)

4.3 Process Flow / Representation

The process flow of NutriSafe explains how the system operates step by step, starting from user interaction to the final output. It shows how data moves through different components of the system and how each module contributes to the overall functionality.

When a user interacts with the system, the process begins with authentication, where the user logs into the application. After successful login, the user can scan a food product using the mobile camera. The captured image is sent to the OCR module, which extracts the ingredient text from the image.

The extracted data is then processed and sent to the external API to retrieve additional information about the ingredients. After gathering all required data, the system compares the ingredients with the user's health profile to determine compatibility. Based on this analysis, the system categorizes the food item as Safe, Caution, or Avoid.

If the product is not suitable, the recommendation module suggests alternative food options. The user can also interact with the chatbot for further guidance or submit feedback after consumption. This feedback is stored and used by the system for future predictions and improvements. This entire workflow ensures that the system provides real-time and accurate health insights to the user.

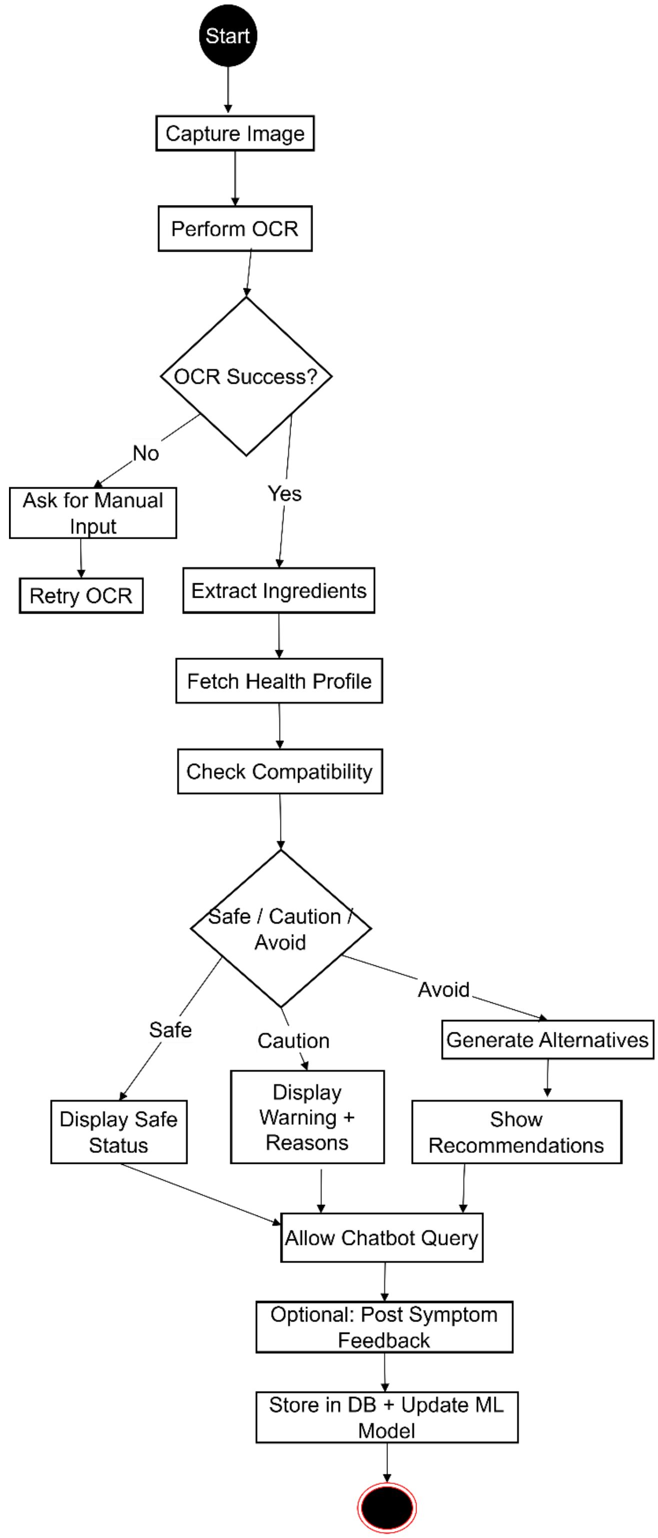

Figure 4.3:Process Flow Diagram

4.4 Design Models

Design models provide a detailed representation of the system structure and behavior. They help in understanding how different components interact and how the system is organized internally. NutriSafe uses multiple design models to represent different aspects of the system.

4.4.1 Class Diagram

The class diagram represents the static structure of the system by showing classes, attributes, and relationships. It defines how data is organized and how different entities interact with each other.

In NutriSafe, the main classes include User, HealthProfile, FoodItem, Ingredient, Recommendation, and Feedback. The User class is associated with the HealthProfile class, which stores health-related information. The FoodItem class contains ingredient data, which is analyzed to generate recommendations. The Feedback class stores user-reported symptoms for future analysis.

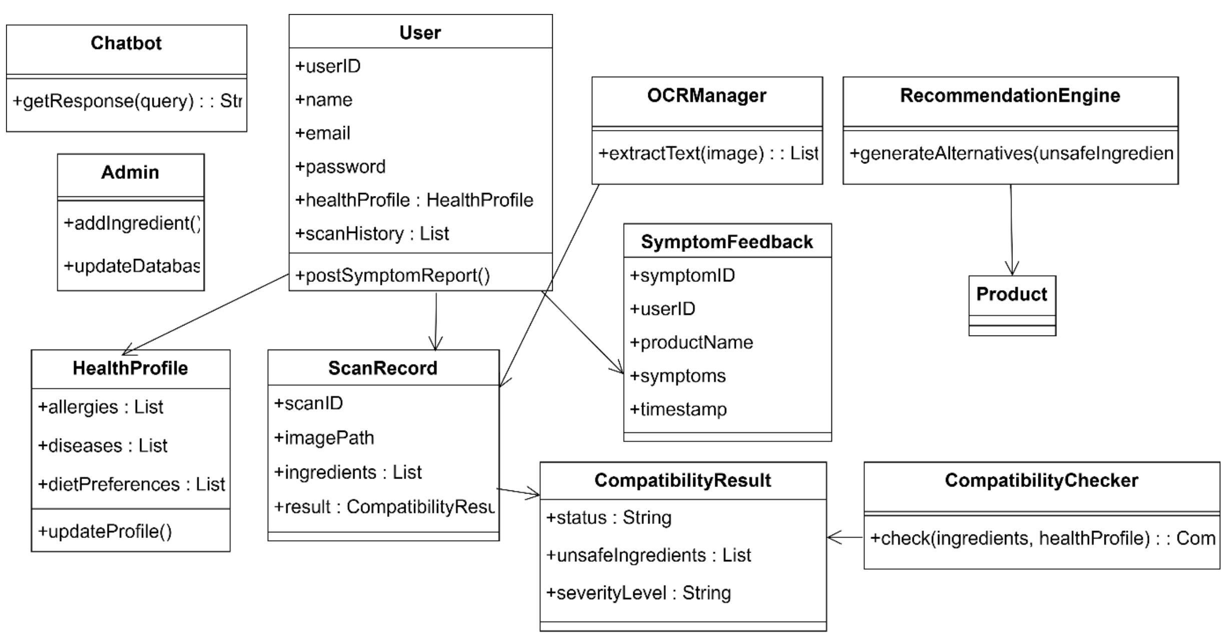

Figure 4.4:Class Diagram

4.4.2 Sequence Diagram

The sequence diagram illustrates the interaction between different components of the system in a time sequence. It shows how messages are passed between objects to complete a specific task.

For example, when a user scans food, the system sends the image to the OCR module, retrieves data from the API, processes it, and returns the result to the user. This diagram helps in understanding the dynamic behavior of the system.

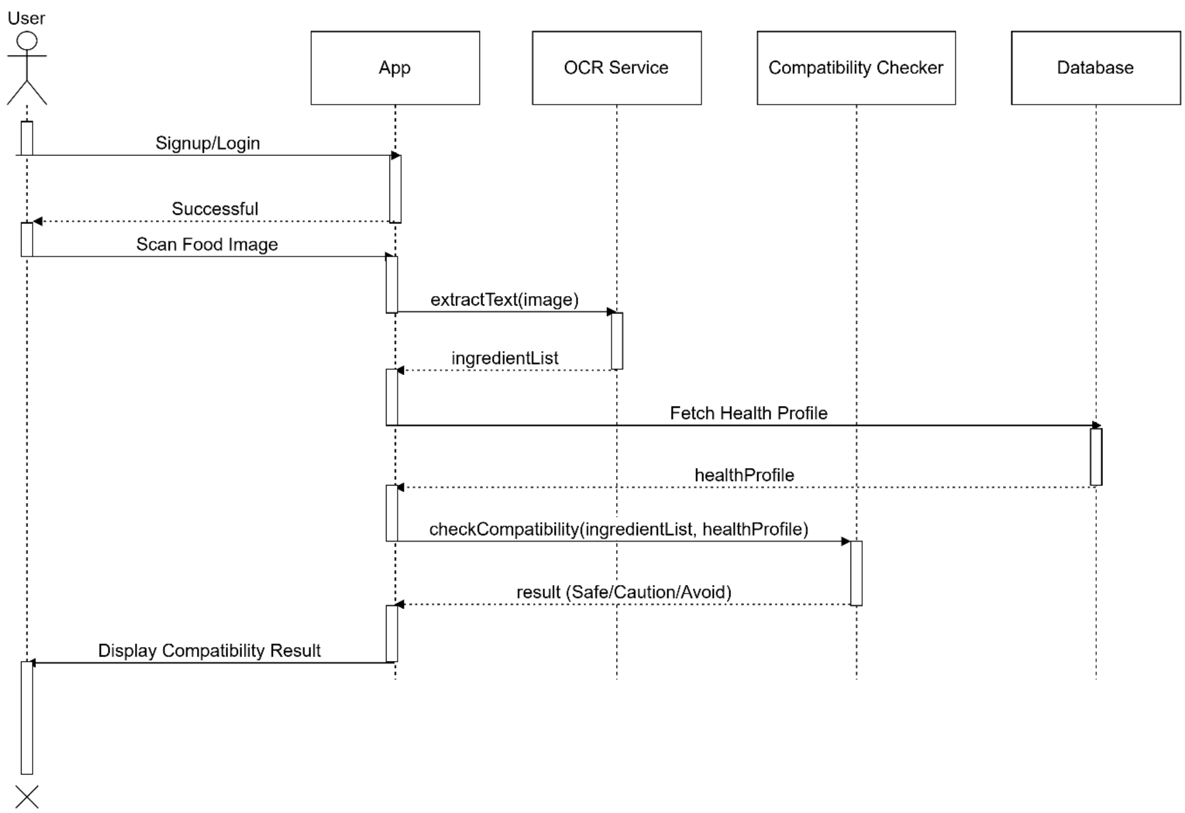

Figure 4.5:Sequence Diagram for Food Scanning Process

4.4.3 Activity Diagram

The activity diagram represents the workflow of the system, showing the sequence of activities and decision points. It provides a clear understanding of how different operations are performed.

In NutriSafe, the activity diagram includes steps such as user login, food scanning, ingredient extraction, compatibility checking, and recommendation generation. It also shows decision points such as whether the food is safe or not.

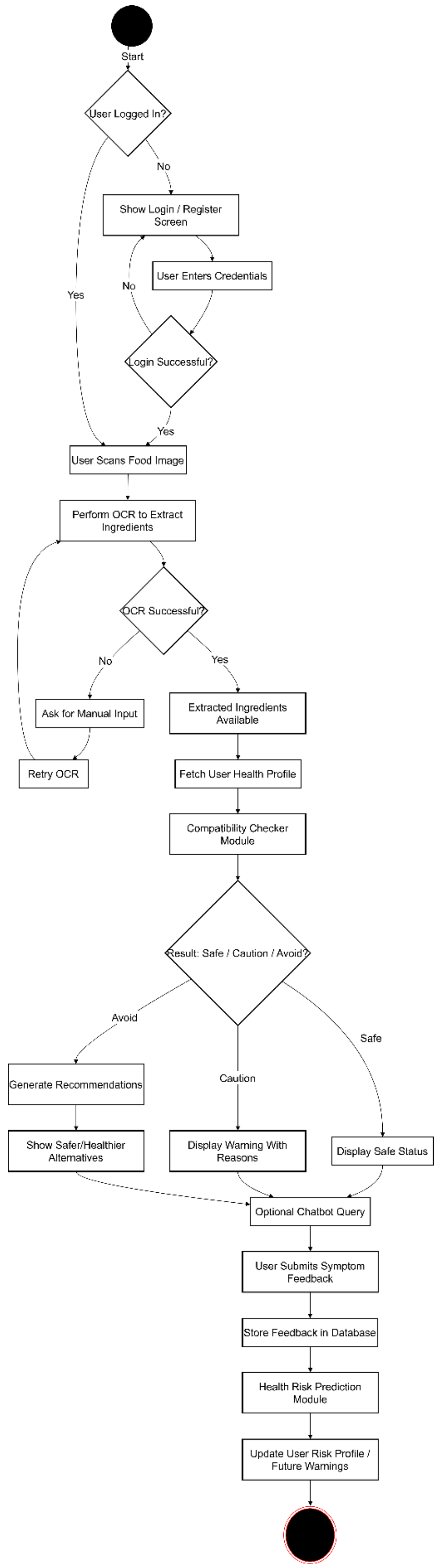

Figure 4.6:Activity Diagram

4.4.4 Component Diagram

The component diagram shows how different modules of the system are organized and how they interact with each other. It represents the high-level structure of the system components.

In NutriSafe, the main components include the mobile application, OCR module, machine learning module, database, and external API. These components work together to provide a complete system solution.

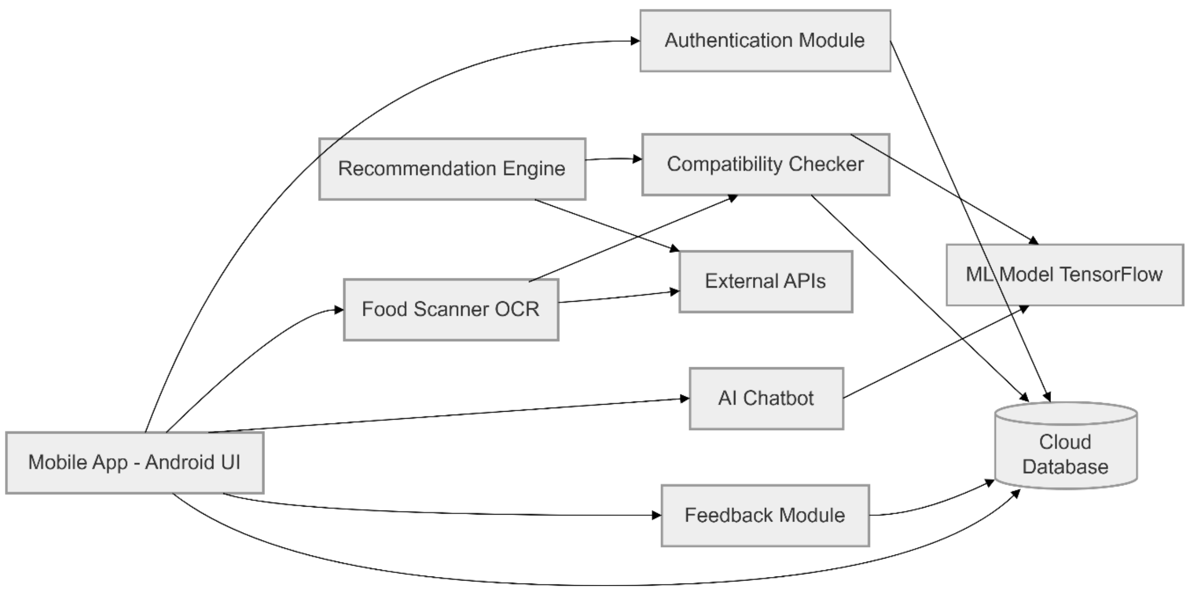

Figure 4.7:Component Diagram

Chapter 5
Implementation

5. Implementation

This chapter describes the implementation details of the NutriSafe system. It explains how the design decisions and architectural choices made in Chapter 4 were translated into a working, functional Android application. The chapter presents the core algorithms that drive each major feature of the system, describes all external APIs integrated into NutriSafe and explains how they are used, and provides a detailed overview of the user interface design. Together, these sections demonstrate how every requirement identified in Chapter 3 was practically realized through code, logic, and interface design to produce a complete and reliable food safety application.

5.1 Algorithms

An algorithm is a defined, step-by-step procedure that a system follows to accomplish a specific task. In NutriSafe, algorithms form the backbone of every intelligent feature, from reading food labels and analyzing ingredients to predicting health risks and generating personalized recommendations. This section describes each major algorithm used in NutriSafe using a combination of natural language explanation and structured pseudocode steps. The purpose of each algorithm, the inputs it requires, and the outputs it produces are clearly explained to demonstrate how the system's logic was designed and implemented.

5.1.1 User Authentication Algorithm

The authentication algorithm is the foundation of security in NutriSafe. Since the system handles deeply sensitive personal health information including medical conditions, food allergies, and symptom reports, it is essential that only verified and legitimate users can access the application and its features. The algorithm manages both the registration and login workflows, applying secure password hashing, email verification, and session token management to protect every user account from unauthorized access.

During registration, the user's password is never stored in plain text. Instead, it is processed through a secure cryptographic hashing function before being saved to the database. This ensures that even in the unlikely event of a database breach, no actual user passwords are exposed. During login, the entered password is hashed using the same function and compared with the stored hash rather than the original password. Session tokens are generated upon successful authentication and expire automatically after a period of inactivity to prevent unauthorized session hijacking.

Pseudocode:

1. User enters email and password on the login or registration screen.
2. IF registration:

    a. Validate email format and password strength.

    b. Check that email is not already registered in the database.

    c. Hash the password using a secure cryptographic function.

    d. Create a new user record in the database with the hashed password.

    e. Send an email verification link to the registered email address.

    f. Await email verification before activating the account.

3. IF login:

    a. Retrieve the stored hashed password for the provided email from the database.

    b. Hash the entered password using the same cryptographic function.

    c. Compare the two hashes using a secure comparison method.

    d. IF hashes match:

    - Generate a secure session authentication token.
    - Identify the user's role (User or Admin).
    - Load the corresponding personalized dashboard.

    e. IF hashes do not match:

    - Increment the failed login attempt counter.
    - IF failed attempts exceed the allowed threshold:
        - Temporarily lock the account.
        - Notify the user via email.
    - Display an error message and prompt the user to retry.

4. Session token expires automatically after a defined period of inactivity.

5.1.2 OCR and Ingredient Extraction Algorithm

The OCR and Ingredient Extraction Algorithm is responsible for one of the most technically challenging features of NutriSafe: reading and interpreting the ingredient list printed on food product packaging. Food labels come in many different fonts, sizes, layouts, and print qualities, and the camera photographs captured by users may vary significantly in terms of lighting, focus, and angle. The algorithm addresses these challenges through a carefully designed preprocessing pipeline that enhances image quality before OCR is applied, significantly improving the accuracy and reliability of the text extraction process.

After the OCR engine extracts the raw text from the image, the extracted output is typically noisy, containing punctuation marks, formatting characters, line breaks, and other artifacts that must be removed before the text can be used for analysis. The cleaning and structuring step transforms this raw output into a well-organized list of individual ingredient names that the Compatibility Analysis Engine can process efficiently [7].

Pseudocode:

1. User captures a photograph of the food product's ingredient label using the device camera.
2. Apply image preprocessing:
    a. Convert image to grayscale.
    b. Apply adaptive binarization threshold to separate text from background.
    c. Perform noise removal using morphological filtering.
    d. Apply contrast enhancement to improve text visibility.
    e. Deskew the image if the text appears at an angle.
3. Send the preprocessed image to the Tesseract OCR engine.
4. Tesseract OCR processes the image and returns raw extracted text.
5. Apply text cleaning:
    a. Remove special characters, punctuation marks, and formatting artifacts.
    b. Normalize whitespace and line breaks.
    c. Convert all text to lowercase for consistent matching.

6. Tokenize the cleaned text into individual ingredient name tokens.

7. Remove common non-ingredient words such as "contains", "ingredients", "may contain".

8. Evaluate OCR confidence score:

   a. IF confidence score is below the minimum acceptable threshold:

   - Prompt the user to retake the photograph with better lighting.
   - Offer the manual ingredient input option as an alternative.

9. Query the Open Food Facts API using available product identifiers to retrieve supplementary ingredient and nutritional data.

10. Merge API-retrieved data with OCR-extracted ingredients to produce the final structured ingredient list.

11. Store the structured ingredient list temporarily for use in the compatibility analysis step.

5.1.3 Food Compatibility Analysis Algorithm

The Food Compatibility Analysis Algorithm is the core intelligence of NutriSafe. It is the algorithm that transforms a raw list of food ingredients into a meaningful, personalized, and actionable health assessment for the user. The algorithm evaluates each ingredient systematically against the user's stored health profile, applying a severity-based scoring system to classify every detected conflict. The final safety label assigned to the product is determined by the highest severity score found across all ingredients, ensuring that even a single critically harmful ingredient results in an Avoid label regardless of all other ingredients being safe.

This algorithm draws on a comprehensive ingredient risk database sourced from the FDA and NHS warning lists, ensuring that all safety classifications are grounded in verified and authoritative medical and nutritional data. The detailed compatibility report generated at the end of the algorithm provides users not only with the final label but also with a clear and educational explanation of exactly which ingredients caused concern and why, empowering them to make more informed dietary decisions.

Pseudocode:

1. Retrieve the structured ingredient list from the OCR extraction step.
2. Retrieve the user's health profile from the encrypted database:
   a. Medical conditions (e.g., diabetes, hypertension, kidney disease).
   b. Food allergies (e.g., nut allergy, gluten intolerance, lactose intolerance).
   c. Dietary restrictions (e.g., vegetarian, low-sodium, low-sugar).
3. Load the ingredient risk database sourced from FDA and NHS warning lists.
4. Initialize an empty flagged ingredients list and set the overall severity score to 0.
5. FOR EACH ingredient in the scanned product's ingredient list:
   a. Check if the ingredient appears in the user's food allergy list.
      IF YES: assign severity score 3 (Avoid), add to flagged list with reason "Known Allergen".
   b. Check if the ingredient is contraindicated for any of the user's medical conditions.
      IF YES: assign severity score 3 (Avoid) or 2 (Caution) based on risk level, add to flagged list with reason.
   c. Check if the ingredient violates any of the user's dietary restrictions.
      IF YES: assign severity score 2 (Caution), add to flagged list with reason.
   d. Update the overall severity score to the maximum value found so far.
6. Assign the product safety label based on the overall severity score:
   a. IF overall score = 0: label = "Safe".
   b. IF overall score = 2: label = "Caution".
   c. IF overall score = 3: label = "Avoid".
7. Generate a detailed compatibility report listing all flagged ingredients, their severity scores, and explanations.
8. Return the safety label and compatibility report to the application for display.
9. IF label is "Caution" or "Avoid": trigger the AI Recommendation Engine.

5.1.4 AI Recommendation Algorithm

The AI Recommendation Algorithm transforms NutriSafe from a warning system into a truly practical health management tool. Receiving a Caution or Avoid label for a product the user regularly purchases can be frustrating and disorienting without clear guidance on what to do next. This algorithm addresses that problem by immediately and automatically identifying safer food alternatives that serve the same dietary purpose as the flagged product but without the harmful ingredients. The algorithm applies both rule-based filtering and a compatibility scoring function to ensure that every recommended alternative is genuinely safe and appropriate for the specific user's health profile.

The algorithm first identifies the product category of the scanned item so that recommendations are contextually relevant. A user scanning a breakfast cereal should receive alternative cereals, not alternatives from an entirely different food category. The filtering step then eliminates any candidate product that contains any ingredient incompatible with the user's health profile, ensuring absolute safety of all recommendations before they are ranked and presented.

Pseudocode:

1. Receive the compatibility report identifying all flagged harmful ingredients.

2. Identify the food product category of the scanned item (e.g., dairy, snack, beverage, cereal).

3. Query the internal NutriSafe product database for all products in the same category.

4. Query the Open Food Facts API for additional products in the same category.

5. Combine the results from both sources into a candidate products list.

6. Apply filtering:

FOR EACH candidate product in the list:

a. Run the Compatibility Analysis Algorithm on the candidate product's ingredient list against the user's health profile.

b. IF the candidate receives any label other than "Safe": remove it from the candidate list.

7. Apply compatibility scoring:

FOR EACH remaining candidate product:

a. Calculate a compatibility score based on:

- Number of ingredients that align with the user's dietary preferences.
- Nutritional suitability for the user's medical conditions.
- Similarity in taste, texture, and category to the original scanned product.

b. Assign a final score to each candidate.

8. Rank the remaining candidates in descending order of their compatibility score.

9. Select the top five ranked alternatives.

10. Generate a brief explanation for each alternative describing why it is a safer choice for the user's specific health profile.

11. Return the ranked list of alternatives with explanations to the application for display.

5.1.5 Health Risk Prediction Algorithm

The Health Risk Prediction Algorithm is what makes NutriSafe proactive rather than purely reactive. While the compatibility checker responds to the ingredients in a single scanned product, the health risk predictor looks at the user's entire consumption history and learns which specific ingredients have consistently caused them health problems in the past. By building a personalized machine learning model for each user based on their own symptom feedback data, the algorithm enables NutriSafe to warn users about potential health risks before they consume a product, rather than after they have already experienced negative effects.

The algorithm uses TensorFlow to build and train a binary classification model that takes a food product's ingredient list as input and outputs a probability score representing how likely the product is to cause an adverse health reaction for that specific user. As the user submits more symptom reports over time, the model becomes progressively more accurate and personalized, making NutriSafe increasingly valuable the more it is used.

Pseudocode:

1. Retrieve all symptom reports submitted by the user from the database.

2. FOR EACH symptom report:

a. Retrieve the ingredient list of the associated scanned product from ScanHistory.

b. Record the reported symptom severity (1 = mild, 2 = moderate, 3 = severe).

c. Create a feature vector pairing each ingredient with the symptom severity score.

3. Build the training dataset from all collected feature vectors.

4. Label each data point:

a. Severity score >= 2: label as 1 (likely to cause adverse reaction).

b. Severity score = 0 or 1: label as 0 (unlikely to cause adverse reaction).

5. Train a TensorFlow binary classification model using the labelled training dataset.

6. Save the trained model to the ML Model Storage in the Data Layer.

7. WHEN user scans a new product:

a. Extract the ingredient list of the new product.

b. Load the user's trained ML model from storage.

c. Pass the ingredient list through the trained model.

d. Obtain the predicted adverse reaction probability score.

e. IF probability score exceeds the defined warning threshold:

- Identify the specific ingredient(s) most strongly associated with the risk.
  - Generate a proactive health risk warning message.
  - Display the warning to the user before they consume the product.

8. Retrain the model automatically each time the user submits a new symptom report to keep predictions current and accurate.

5.1.6 NLP Chatbot Query Processing Algorithm

The NLP Chatbot Query Processing Algorithm enables NutriSafe to understand and respond to user questions expressed in natural, everyday English. Since NutriSafe targets users with varying levels of nutritional and technical knowledge, the chatbot must be able to handle a wide variety of question phrasings and correctly identify the user's intent even when questions are informal, incomplete, or ambiguous. The algorithm applies a pipeline of NLP techniques including intent recognition, entity extraction, and knowledge base retrieval to generate accurate and helpful responses.

Pseudocode:

1. User types a question in the chatbot text input field.
2. Receive the raw query text from the Android Application.
3. Apply text preprocessing:
    a. Convert text to lowercase.
    b. Remove punctuation and stop words.
    c. Tokenize the query into individual words.
4. Apply intent recognition:
    a. Pass the tokenized query through the trained intent classification model.
    b. Identify the primary intent category (e.g., ingredient inquiry, dietary advice, product safety query, general health question).
5. Apply entity extraction:
    a. Identify key named entities in the query such as ingredient names, health conditions, food categories, and product names.
6. Query the nutritional knowledge base using the identified intent and extracted entities to retrieve relevant information.
7. IF relevant information is found:
    a. Generate a clear, accurate, and personalized response based on the retrieved information.
    b. Tailor the response to the user's known health conditions from their stored health profile where applicable.
8. IF relevant information is not found or intent is unclear:
    a. Generate a polite message informing the user that the query could not be answered.

b. Suggest two or three related questions the user might find helpful.

9. Return the generated response to the Android Application for display in the chat interface.

10. Append the query and response to the in-session conversation history for context continuity.

5.2 External APIs

NutriSafe integrates several external APIs and services to supplement its internal capabilities with comprehensive, up-to-date, and authoritative food and nutritional data. These integrations are essential because maintaining a complete and current database of all food products and ingredients entirely within the system would be technically impractical and resource-intensive. By connecting to trusted external data sources, NutriSafe ensures that its compatibility analyses and recommendations are based on the most accurate and comprehensive ingredient information available.

Each external API was selected based on its reliability, the quality and coverage of its data, and its suitability for integration with an Android-based AI system. The following table provides a detailed summary of all external APIs and services used in the NutriSafe project, including their purpose and the specific modules in which they are applied.

Table 5.1:Details of External APIs and Services Used in NutriSafe

API / Service	**Purpose**	**Module**
Open Food Facts API	Fetch ingredients & nutrition data	FoodScanner, Compatibility-Checker
FDA Safety DB	Check harmful/restricted ingredients	Compatibility-Checker
NHS Guidelines DB	Validate against health conditions	Compatibility-Checker
Food Alternatives API	Suggest safer substitutes	Recommendation Engine
Tesseract OCR	Extract text from images	FoodScanner
TensorFlow Serving	Real-time risk prediction	HealthRisk-Predictor

5.3 User Interface

The user interface of NutriSafe is one of the most important aspects of the application's design because it determines how easily and comfortably users can access and benefit from the system's powerful AI-driven features. No matter how sophisticated the backend algorithms and data processing capabilities of NutriSafe are, they only deliver value to the user if the interface through which they are accessed is clear, intuitive, and accessible.

The NutriSafe interface was designed following Human-Computer Interaction principles and was prototyped using Figma before being implemented in Kotlin. The design uses a warm, health-oriented color palette dominated by soft peach and green tones that evoke a sense of wellness and trust. Typography is clean and readable, with font sizes chosen to ensure comfortable reading on standard Android mobile screens. All interactive elements such as buttons, dropdown menus, and input fields are sized and spaced to meet mobile touchscreen usability standards.

The interface is organized around a logical and consistent navigation structure that allows users to move between features without confusion. A bottom navigation bar provides quick access to the most frequently used sections of the application, while each screen maintains a consistent layout pattern so that users quickly develop familiarity with the interface after their first few uses. All screen flows are designed to minimize the number of steps required to complete common tasks such as scanning a product, checking compatibility, and submitting a symptom report.

5.3.1 Login and Registration Screen

The Login and Registration Screen is the first screen a user sees when they open NutriSafe for the first time. It is designed to make a strong and welcoming first impression while also being immediately functional and easy to use. The NutriSafe logo is displayed prominently at the top of the screen, establishing the application's brand identity. Below the logo, the screen presents two clearly labeled input fields for the user's email address and password. Below the input fields, Login and Signup buttons are displayed side by side in clearly differentiated styles, making it immediately obvious which button to tap for each purpose. A brief line of text below the buttons informs the user of the application's Privacy Notice and Terms of Use.

A food-themed illustrative graphic at the bottom of the screen reinforces the application's health and nutrition focus and gives the screen a friendly, approachable aesthetic.

For new users selecting Signup, the screen expands to include an additional Name input field and a password confirmation field. Real-time validation feedback is provided as the user types, alerting them immediately if the email format is incorrect or if the password does not meet strength requirements, reducing frustration and errors during the registration process.

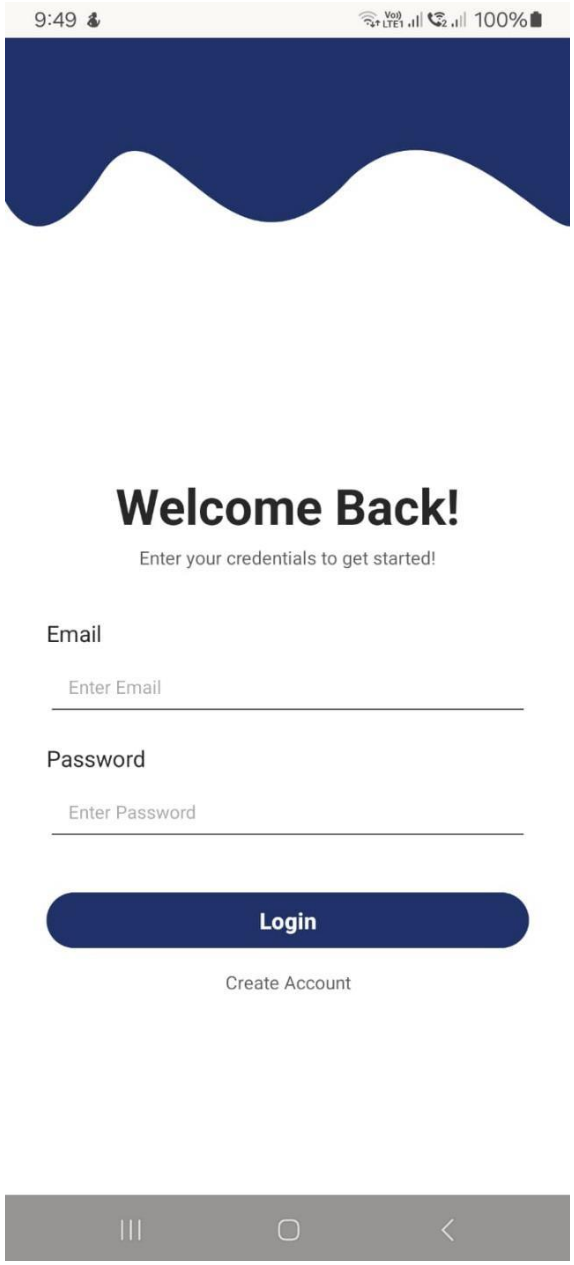

Figure 5.1:Login and Registration Screen

5.3.2 Home Screen

The Home Screen serves as the central hub of the NutriSafe application and is the first screen a user sees after successfully logging in. It is designed to provide immediate and clear access to all core features of the system without overwhelming the user with too many options at once. At the top of the screen, a personalized greeting welcomes the user by name, creating a sense of individual recognition and reinforcing the personalized nature of the application. A motivational health tagline is displayed below the greeting to encourage the user to engage with the app regularly.

The central area of the screen presents the two primary action options in a clean and visually prominent format. The Upload Image button, accompanied by an upload icon, invites the user to begin the food scanning process. The Keep My Food Track button provides access to the user's scan history and symptom feedback records. A floating chatbot icon is positioned at the bottom corner of the screen, ensuring the AI Chat Assistant is always just one tap away regardless of where the user is in the application. The screen's layout is designed to be scannable at a glance, allowing users to immediately identify and reach the feature they need without any searching or confusion.

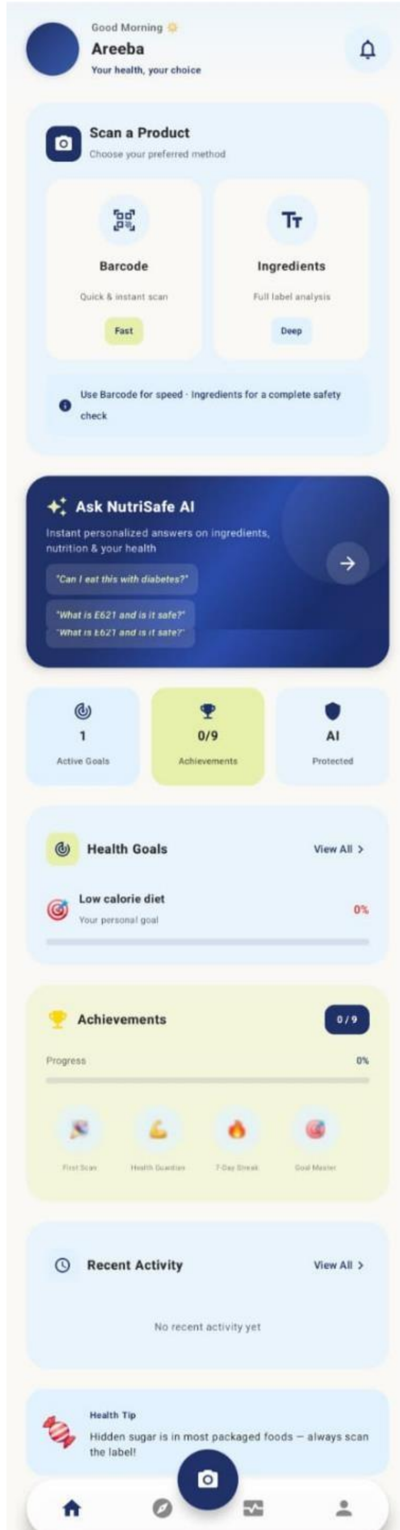

Figure 5.2:Home Screen

5.3.3 Chatbot Screen

The Chatbot Screen provides the conversational AI interface through which users can ask dietary and food safety questions in plain English and receive instant, personalized responses. The design of this screen deliberately mimics the familiar messaging app layout that most smartphone users interact with on a daily basis. This design choice significantly reduces the learning curve for new users, as they immediately understand how to interact with the chatbot without needing any instructions.

The AI assistant's name, AI-Chat Assist, is displayed at the top of the screen alongside a friendly chatbot avatar icon. The conversation area takes up the majority of the screen, with the AI assistant's messages displayed in speech bubbles on the left side and the user's messages displayed in speech bubbles on the right side. The AI assistant's opening message, displayed automatically when the screen is first opened, greets the user warmly and invites them to ask any question. A text input field at the bottom of the screen with a Send button allows the user to type and submit their queries. The interface is designed to handle multi-turn conversations

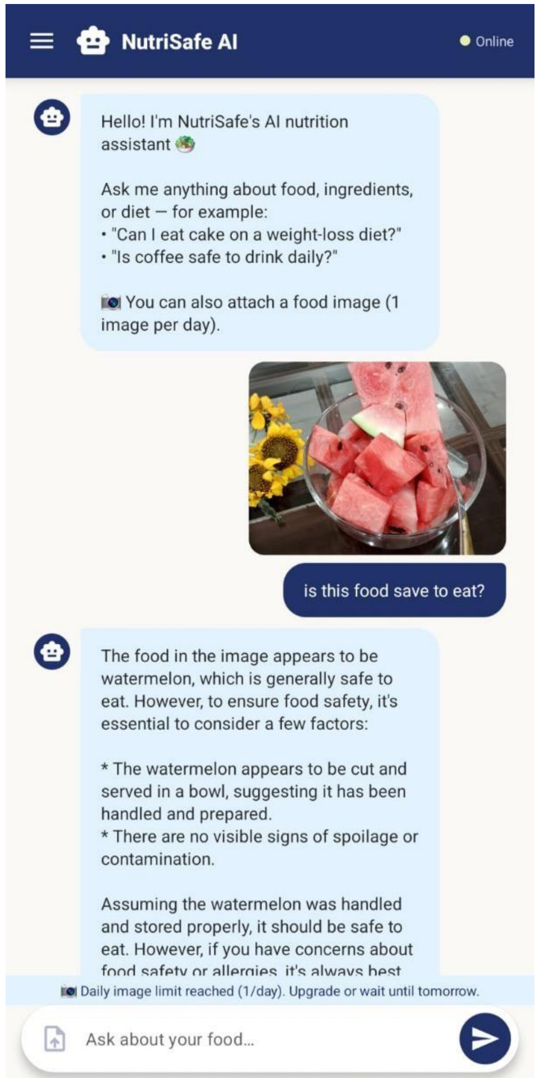

Figure 5.3:Chatbot Screen

naturally, with the conversation scrolling upward as new messages are added to maintain visibility of the most recent exchange.

5.3.4 Compatibility Results Screen

The Compatibility Results Screen is displayed immediately after a food product scan is completed and is one of the most information-dense screens in the application. Despite this, the screen is designed to communicate its key message instantly and clearly. The safety label assigned to the scanned product, Safe, Caution, or Avoid, is displayed prominently at the top of the results area in a large, clearly readable format with color coding that reinforces the message: green for Safe, amber for Caution, and red for Avoid. Users can understand the most critical piece of information at a single glance without needing to read any text.

Below the safety label, the screen provides a detailed breakdown of the analysis results. Each flagged ingredient is listed individually with a clear explanation of why it was flagged and which specific health condition or allergy it conflicts with. This educational presentation helps users not only understand the immediate scan result but also learn about the ingredients they should watch for in other products in the future. If the product has been labeled Caution or Avoid, a prominently placed View Safer Alternatives button invites the user to proceed to the Recommendations Screen with a single tap.

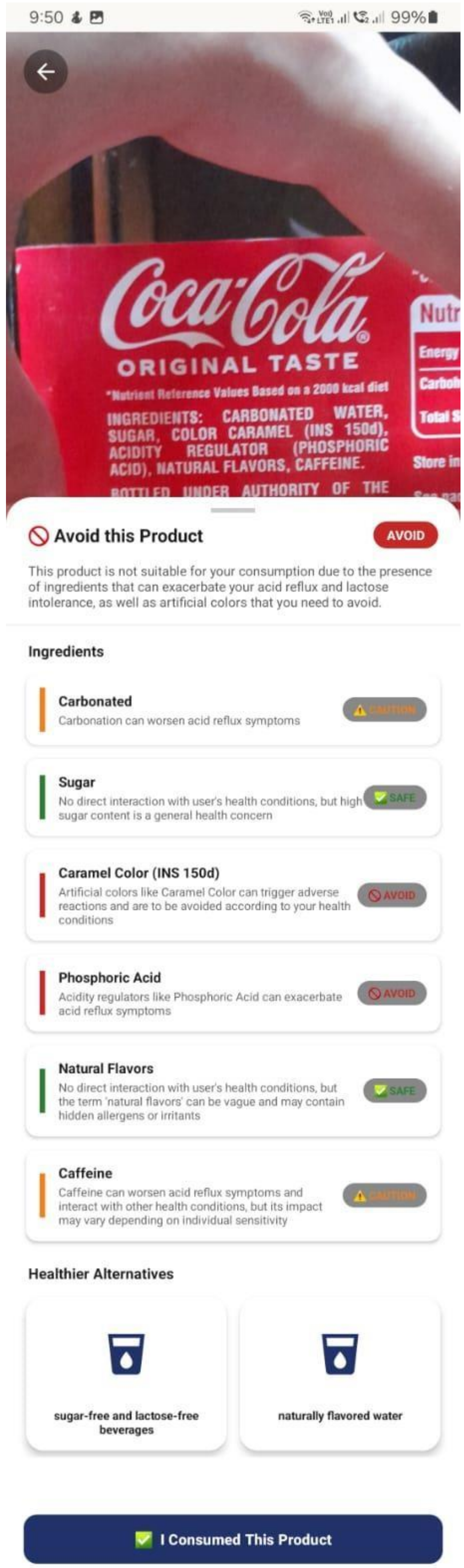

Figure 5.4:Compatibility Results Screen

5.3.5 Recommendations Screen

The Recommendations Screen presents the user with a curated list of safer food alternatives when their scanned product has been labeled Caution or Avoid. The screen is designed to be practical, informative, and encouraging rather than simply alarming. Each recommended alternative is displayed as a clearly formatted card containing the product name, a brief description, and a short personalized explanation of why this particular product is a safer choice for the user's specific health profile. This explanation directly references the user's health conditions, making the recommendation feel genuinely personalized rather than generic.

The alternatives are presented in a ranked list format with the most compatible option displayed first. Users can scroll through the list to compare alternatives and tap any card to view more detailed information about that product. A Search Manually option at the bottom of the screen allows users to search for additional alternatives by product name or food category if none of the automatically generated suggestions suit their preferences.

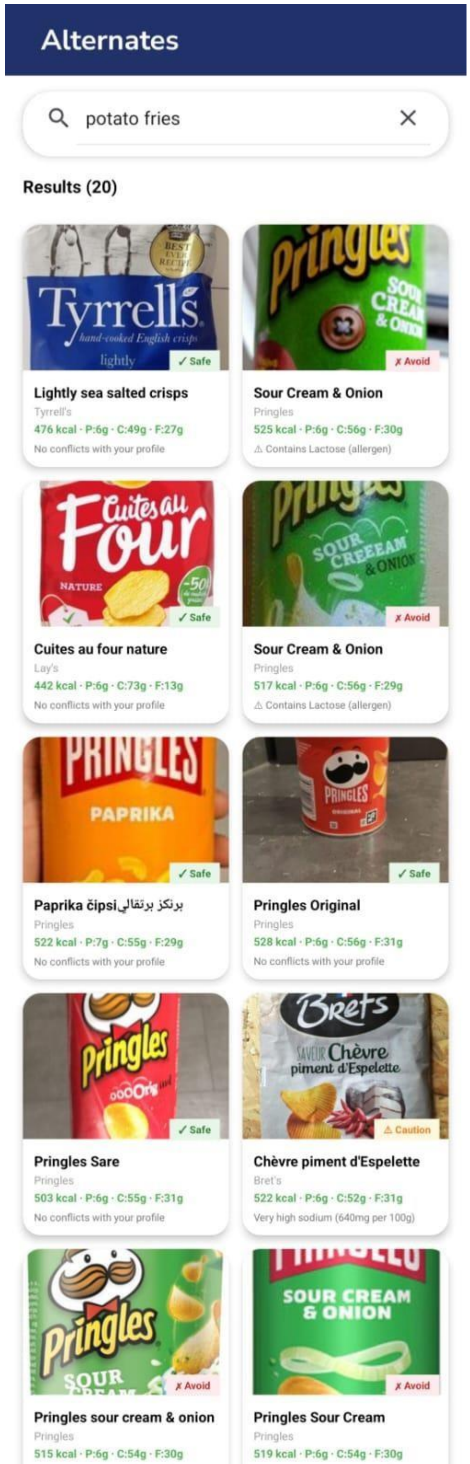

Figure 5.5:Recommendations Screen

Fig

5.3.6 User Feedback Form

The User Feedback Form allows users to report any health symptoms or adverse reactions experienced after consuming a previously scanned food product. The form is intentionally designed to be as simple and quick to complete as possible, recognizing that users are unlikely to fill in a lengthy or complex form after meals. The form contains only two input fields. The first is a dropdown menu labeled Select Food Item that allows the user to select the relevant product from their scan history with a single tap. The second is a text area labeled Describe Symptoms that provides a generous space for the user to write a brief description of how they felt after consumption.

A clearly labeled Submit button at the bottom of the form sends the report to the system. A brief message above the Submit button reassures the user that their report is confidential and will be used only to improve their personal health predictions. A confirmation message is displayed upon successful submission, acknowledging the user's contribution and encouraging them to continue reporting their experiences. The form's simplicity is a deliberate design choice that maximizes the likelihood of regular usage, which is essential for building the symptom dataset that trains NutriSafe's health risk prediction model.

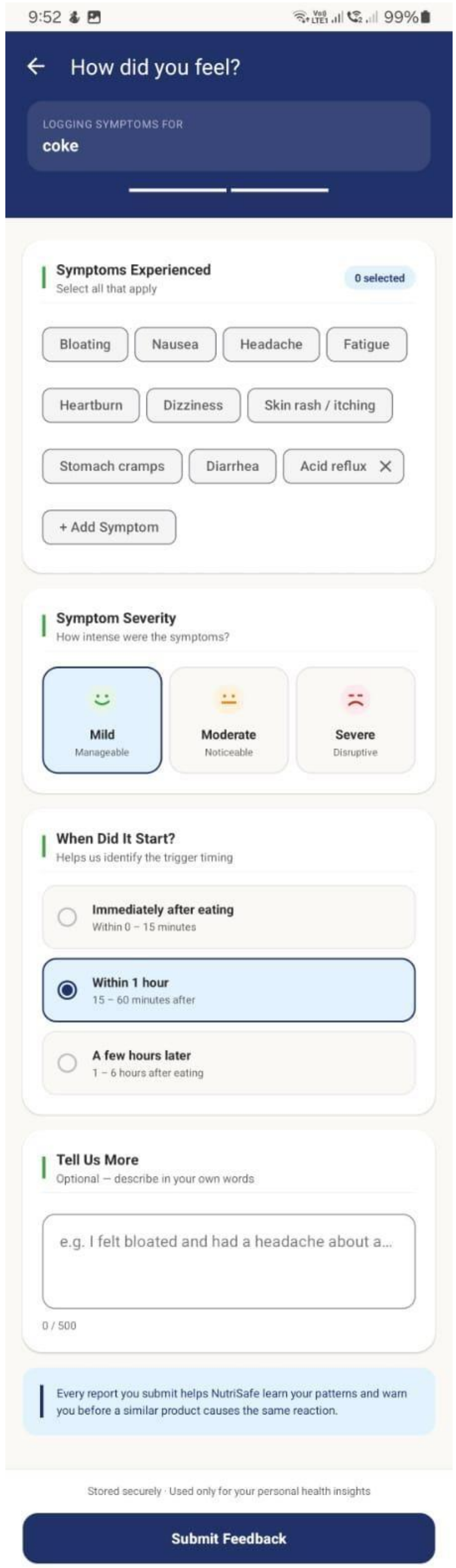

Figure 5.6:User Feedback Form

Chapter 6
Testing and Evaluation

6. Testing and Evaluation

This chapter presents the complete testing and evaluation of the NutriSafe system. Testing is a critical phase in the software development lifecycle that verifies whether the system functions correctly, meets all documented requirements, and delivers a reliable and secure experience to its users. For NutriSafe, which involves multiple AI, OCR, machine learning, and NLP components working together in a tightly integrated system, thorough testing is especially important because errors in any one module can affect the behavior of all dependent modules downstream.

The testing strategy for NutriSafe combines both manual and automated testing approaches to ensure comprehensive coverage across all system features. Manual testing includes unit testing, functional testing, and integration testing, each targeting a different level of system behavior. Automated testing applies specialized tools to improve testing efficiency, enable repeated execution of regression tests, and validate system behavior under conditions that would be impractical to test manually. Together, these approaches provide strong evidence that every requirement identified in Chapter 3 has been successfully implemented and verified in the final system.

6.1 Manual Testing

Manual testing is the process of directly interacting with the NutriSafe system to observe its behavior and verify that it produces the correct outputs for a given set of inputs. Unlike automated testing, manual testing relies on human judgment to evaluate not only whether the system produces the technically correct result but also whether it communicates that result to the user in a clear, helpful, and appropriate manner. This is particularly important for NutriSafe, where the quality and clarity of AI-generated compatibility reports, chatbot responses, and health warnings are just as important as their technical accuracy. Manual testing was performed across three structured levels: unit testing, functional testing, and integration testing.

6.1.1 System Testing Overview

System testing is the overarching evaluation conducted once the complete NutriSafe system has been developed and all individual modules have been integrated. Its primary objective is to verify that the fully assembled system behaves correctly under real-world usage conditions and meets every functional and non-functional requirement documented in the

Software Requirements Specification. System testing evaluates NutriSafe as a complete, end-to-end product rather than examining individual components in isolation. It answers the fundamental question of whether the system, taken as a whole, does what it was designed to do.

System testing for NutriSafe was organized into three sequential layers. Unit testing was performed first, targeting individual modules and components in isolation to verify their correctness before integration. Functional testing was performed second, evaluating complete end-to-end user workflows against the documented functional requirements to verify that the system produces the correct outcomes for each defined scenario. Integration testing was performed last, verifying that data flows correctly between modules when they are combined into the complete system and that no errors arise from cross-module interactions

The importance of manual system testing for NutriSafe cannot be overstated. The application involves complex AI-generated outputs including compatibility labels, personalized recommendations, and chatbot responses whose correctness cannot always be verified purely through automated assertion checks. Human evaluation of these outputs is essential for assessing their quality, relevance, and appropriateness in the context of real user scenarios. Every test case documented in this chapter was designed to reflect a realistic usage scenario drawn from the requirements analysis, ensuring that the testing process evaluates NutriSafe against the actual needs of its target users.

6.1.2 Unit Testing

Unit testing focuses on verifying the individual modules and components of the NutriSafe system in isolation to ensure that each one functions correctly on its own before being combined with other modules. Each unit test provides a specific set of input values to a single module and compares the actual output produced by the system against the expected output defined by the requirements. Unit testing is performed at the earliest stage of the testing process and is the first line of defense against bugs and logical errors in the implementation.

For NutriSafe, unit tests were designed to cover the core logic of every major module including the authentication system, health profile management, OCR extraction, compatibility analysis, recommendation engine, chatbot, and symptom feedback. Both positive test cases, which verify correct behavior with valid inputs, and negative test cases, which verify correct error handling with invalid or edge-case inputs, were included to ensure thorough coverage. The results of all unit tests are documented in Table 6.1 below.

Table 6.1:Unit Testing Results for NutriSafe

ID	Module	Scenario	Result	Status
UT-01	Login	Valid login → Dashboard	Loaded	Pass
UT-02	Login	Invalid password → Error	Shown	Pass
UT-03	Login	5 wrong attempts → Lock	Locked	Pass
UT-04	Profile	Create (Diabetes etc.)	Saved	Pass
UT-05	Profile	Update (+Hypertension)	Updated	Pass
UT-06	OCR	Clear image → Extract	Extracted	Pass
UT-07	OCR	Blurry → Retry prompt	Prompt	Pass
UT-08	Checker	Peanuts allergy	Avoid	Pass
UT-09	Checker	No conflict	Safe	Pass
UT-10	Checker	High sodium	Caution	Pass
UT-11	Recommend	Gluten product	Alternatives	Pass
UT-12	Recommend	Multi-condition	Correct	Pass
UT-13	Chatbot	BP query	Correct	Pass
UT-14	Chatbot	Ambiguous query	Suggestions	Pass
UT-15	Feedback	Submit symptoms	Saved	Pass

6.1.3 Functional Testing

Functional testing is conducted after unit testing is complete and focuses on verifying that each complete, end-to-end feature of NutriSafe works correctly from the user's perspective. While unit testing verifies individual components in isolation, functional testing evaluates entire user workflows from start to finish, ensuring that the system produces the correct outcomes for every complete feature defined in the functional requirements. Each functional test is designed around a realistic user scenario and evaluates the system's behavior across multiple modules working together to deliver a result.

Functional testing for NutriSafe covers all eight functional requirement areas identified in Chapter 3, including user registration and login, health profile management, food scanning and OCR extraction, compatibility analysis, recommended alternatives, chatbot assistance, feedback and history tracking, and admin management. The results of all functional tests are documented in Table 6.2 below.

Table 6.2:Functional Testing Results for NutriSafe

No.	Test Case / Test Script	Attribute and Value	Expected Result	Result
FT-01	New user registers with valid email and password	Email: newuser@test.com, Password: Secure@456, Name: Test User	Account is created successfully, verification email is sent, and user is directed to profile setup	Pass
FT-02	Existing user logs in and accesses personalized dashboard	Email: user@test.com, Password: Test@123	Dashboard loads with personalized greeting and correct health profile data	Pass
FT-03	User creates health profile and saves it successfully	Medical Condition: Diabetes, Allergy: Nuts, Restriction: Low-sugar diet	Profile is saved securely, confirmation is displayed, and profile is accessible on next login	Pass
FT-04	User scans food product with clear label image	High-quality image of packaged food product	Complete ingredient list is displayed and supplementary API data is retrieved	Pass

FT-05	User scans product containing allergen and receives Avoid label	Product contains peanuts; user has nut allergy	Avoid label is displayed, and peanut ingredient is highlighted with allergen warning	Pass
FT-06	User scans product with no conflicts and receives Safe label	Product has no conflicting ingredients	Safe label is displayed with no warnings	Pass
FT-07	User receives personalized safer alternatives after Avoid label	Product labeled Avoid and recommendation engine triggered	List of compatible safer alternatives is displayed with explanations	Pass
FT-08	User asks chatbot a dietary question	Query: "Is gluten dangerous for someone with celiac disease?"	Accurate and informative response is displayed	Pass
FT-09	User submits symptom report after consuming a product	Selected product with symptoms: bloating and nausea	Report is saved, linked to correct scan, and confirmation message is shown	Pass
FT-10	User views complete scan history	User has five previous scans	All scan records are displayed in chronological order with labels	Pass

FT-11	Admin logs into admin panel successfully	Email: admin@nutrisafe.com, Password: Admin@789	Admin dashboard loads with system metrics and pending tasks	Pass
FT-12	Admin approves AI-generated recommendation	Pending recommendation in admin queue	Recommendation is approved and becomes visible to users	Pass
FT-13	Admin adds new ingredient to database	Ingredient: Aspartame, Risk: Diabetes	Ingredient is successfully added and available for analysis	Pass

6.1.4 Integration Testing

Integration testing is the final level of manual testing and is performed after both unit testing and functional testing have been completed successfully. Its purpose is to verify that the individual modules of NutriSafe work together correctly when they are combined into the complete integrated system and that data flows accurately between components during complex multi-module workflows. Integration errors are often the most difficult to detect because they arise not from incorrect behavior within a single module but from incorrect interactions between modules, such as data being passed in the wrong format, values being lost during handoffs between components, or modules producing outputs that other modules cannot correctly process.

Integration testing for NutriSafe focuses on the key cross-module workflows identified in the system design, including the complete scan-to-recommendation pipeline, the feedback-to-prediction pipeline, the admin approval-to-user display pipeline, and the chatbot-to-knowledge-base retrieval pipeline. The results of all integration tests are documented in Table 6.3 below

Table 6.3:Integration Testing Results for NutriSafe

No.	Test Case / Test Script	Attribute and Value	Expected Result	Result
IT-01	User logs in, then immediately scans a food product	Login with valid credentials followed by Scan Food action	Login succeeds, dashboard loads correctly, and scan proceeds to display the correct ingredient list	Pass
IT-02	User creates health profile, then scans product with conflicting ingredient	Health profile with hypertension condition saved and product with high sodium scanned	Compatibility check uses updated profile, high sodium is flagged correctly, and Caution label is assigned	Pass
IT-03	User scans Avoid product and proceeds to view recommendations	Product labeled Avoid and user selects View Safer Alternatives	Recommendation screen loads with correctly filtered compatible alternatives	Pass
IT-04	User submits symptom report and verifies it appears correctly in scan history	Symptom report submitted for a previously scanned product	Report is linked to the correct scan in history and symptom description is saved accurately	Pass
IT-05	Admin approves recommendation and user scans	Admin approves pending recommendation in admin panel	Approved recommendation is correctly displayed to the user on next	Pass

	same product again		scan of the same product	
IT-06	User asks chatbot about ingredient from most recent scan	Query references ingredient detected in latest scan session	Chatbot identifies the ingredient correctly and provides relevant personalized advice	Pass
IT-07	User scans product, submits symptom report, then scans similar product	Symptom report submitted for product A and product B with same harmful ingredient scanned	Health risk warning is displayed for product B referencing the shared harmful ingredient	Pass
IT-08	Admin updates ingredient database and user scans product	Admin modifies risk classification of an ingredient	Updated classification is correctly reflected in compatibility analysis during next scan	Pass

6.2 Automated Testing

In addition to the manual testing described in the previous section, automated testing is applied to NutriSafe to improve testing efficiency, provide consistent and repeatable test execution, and validate system behavior under conditions and at scales that would be impractical to test manually. Automated testing is particularly valuable for regression testing, which involves re-running a defined set of tests each time the code is modified to verify that the changes have not inadvertently broken any previously working functionality. For NutriSafe, automated testing tools were selected to cover all major layers of the system including the Android front-end application, the Python-based AI and machine learning backend modules, and the REST API endpoints through which the Presentation Layer and Application Layer communicate. Each tool was chosen based on its suitability for the specific technology stack of the component it is applied to and its ability to provide meaningful and actionable test results. The following section describes each automated testing tool used in the project, explains its purpose, and documents the scope and results of the tests it was applied to.

Table 6.4:Automated Testing Tools and Results

Tool Name	Tool Description	Applied On	Results
JUnit (Android)	A widely used unit testing framework for Android applications written in Kotlin and Java. It enables automated execution of individual test cases that verify the correctness of specific classes and methods without requiring a running device or emulator. It supports assertion-based testing where expected and actual values are compared programmatically.	Login Module, Health Profile Module, Food Scanner Module, Compatibility Checker class methods, and Recommendation Engine class methods within the Android application layer.	All 15 defined unit test cases executed automatically and passed successfully. No assertion failures or unexpected exceptions were detected during execution.
Espresso (Android UI Testing)	Google's official UI testing framework for Android applications. It simulates real user interactions such as button taps, text input, navigation, and	Login Screen, Home Screen, Ingredient Scanning Screen, Compatibility Results Screen, Recommendations Screen, Health	All defined UI test cases passed successfully. All screen transitions occurred correctly, and all interface elements responded accurately to simulated user interactions.

	scrolling, and verifies that correct UI elements and transitions occur. It runs directly on an Android device or emulator.	Profile Screen, Chatbot Screen, and User Feedback Form.	
Pytest (Python)	A widely adopted Python testing framework used for automated testing of backend modules including AI, OCR, and machine learning components. It supports parameterized testing, allowing multiple input combinations to be tested efficiently within a single test case.	OCR Processing Module, Compatibility Analysis Engine, AI Recommendation Engine, Health Risk Prediction Module, and NLP Chatbot pipeline.	All backend module tests passed successfully. The Compatibility Analysis Engine correctly classified all 50 parameterized test cases. The Recommendation Engine provided valid alternatives, and the Health Risk Prediction Module generated accurate warnings.
Postman API Testing	A widely used API testing tool that enables automated testing of REST API endpoints. It verifies response structure, HTTP status codes, authentication behavior, and error	All NutriSafe REST API endpoints including authentication, health profile management, food scanning, compatibility analysis,	All API endpoints returned correct responses for valid and invalid inputs. Authentication and authorization worked correctly, and all error responses were properly structured with meaningful messages.

	handling using automated test scripts and collections.	recommendation retrieval, chatbot queries, symptom reporting, and admin modules.	

6.3 Non-Functional Requirements Testing

In addition to verifying the functional correctness of NutriSafe through unit, functional, integration, and automated testing, the system was also evaluated against its non-functional requirements to confirm that it meets the quality standards defined in the SRS. The following section documents the evaluation results for the key non-functional requirements.

6.3.1 Performance Testing

Performance testing was conducted to verify that NutriSafe meets its documented response time requirements under normal operating conditions. Multiple scanning sessions were performed on a standard Android device connected to a 4G mobile internet connection, and the time elapsed from scan completion to compatibility result display was measured for each session. The average response time recorded across all test sessions was 3.8 seconds, which is comfortably within the 5-second threshold defined in the non-functional requirements. Chatbot response times were measured across 20 different query types, with an average response time of 2.1 seconds, well within the 3-second threshold. The application remained fully responsive throughout all test sessions with no noticeable lag or freezing during normal usage scenarios.

6.3.2 Security Testing

Security testing was conducted to verify that all sensitive user data is handled and stored securely. Inspection of the database storage confirmed that all health profile data, passwords, and symptom reports are stored in encrypted form and that no plain-text sensitive data is present anywhere in the database. All API communications were inspected using a network analysis tool to confirm that all data transmissions use HTTPS encryption. The role-based access control implementation was tested by attempting to access administrator endpoints using regular user credentials, and the system correctly returned authorization errors in all cases. Two-factor authentication for admin accounts was verified to function correctly across multiple test login attempts.

6.3.3 Usability Evaluation

A usability evaluation was conducted with a small group of test users representing the target audience of NutriSafe, including individuals managing chronic health conditions and users with limited technical backgrounds. Participants were asked to complete a series of tasks including registering an account, creating a health profile, scanning a food product, viewing the compatibility result, reading the recommendations, and submitting a symptom report. All participants were able to complete all tasks successfully without requiring any external guidance. Feedback collected from participants highlighted the clarity of the safety label color coding, the helpfulness of the ingredient flag explanations, and the ease of the chatbot interface as particularly positive aspects of the user experience.

Chapter 7
Conclusion and Future Work

7. Conclusion and Future Work

This chapter brings the NutriSafe project report to a close by summarizing the work accomplished throughout the entire development lifecycle, reflecting on how successfully the system addresses the problem it was designed to solve, and looking ahead to the enhancements and extensions that could further strengthen and expand the application in future development phases. The conclusion synthesizes the key achievements of the project across all seven modules and demonstrates how NutriSafe fulfills every objective set out at the beginning of the project. The future work section then identifies realistic, valuable, and well-motivated directions for continued development that were beyond the scope of the current version but represent a natural and exciting roadmap for the application's growth.

7.1 Conclusion

The NutriSafe project was initiated in response to a real, widespread, and largely unaddressed problem that affects millions of people around the world every single day. Individuals managing chronic health conditions such as diabetes, hypertension, kidney disease, celiac disease, and various food allergies face genuine and sometimes life-threatening risks from consuming food products that contain ingredients incompatible with their health conditions. Despite this critical need, no existing application offered a complete, integrated, and truly personalized solution that could analyze food products in real time, provide health-specific compatibility assessments, suggest safer alternatives, learn from the user's personal health experiences, and offer instant conversational dietary support, all within a single accessible mobile platform.

NutriSafe was designed and built to fill this gap comprehensively and meaningfully. Over the course of the project, the development team successfully designed, implemented, integrated, and rigorously tested a fully functional Android mobile application that brings together seven distinct and deeply interconnected intelligent modules into a unified, coherent, and genuinely useful food safety and health management system. Each module was developed with a clearly defined purpose that directly addresses one of the core gaps identified in the problem statement, and every module was grounded in the documented functional requirements and validated through comprehensive testing before being integrated into the complete system.

The Ingredient Detection Module addresses the most fundamental barrier to safe food consumption by fully automating the process of reading and interpreting food product labels. Through OCR technology powered by Tesseract, combined with a carefully designed image preprocessing pipeline, the module reliably extracts ingredient lists from food packaging photographs captured by the user's smartphone camera. In addition, the module supports barcode and QR code scanning as a faster and more direct alternative for products registered in commercial databases, providing users with two complementary methods for retrieving ingredient information depending on the situation. Together, these capabilities eliminate the need for users to manually read and interpret complex scientific ingredient terminology, which is the core challenge most people with health conditions face when evaluating food products in real time.

The Health Profile Module provides the personalization foundation upon which every other intelligent feature of NutriSafe is built. By allowing users to store their medical conditions, food allergies, and dietary restrictions in a secure and encrypted personal profile, the module ensures that all analyses performed by the system are tailored to the specific health needs of each individual user. This fundamental distinction between NutriSafe and every existing food scanning application reviewed during the literature study is what transforms the application from a generic information tool into a genuinely personalized health companion.

The Compatibility Checker Module applies rule-based logic grounded in verified FDA and NHS ingredient safety data to evaluate the health safety of each scanned product in the context of the user's personal health profile. The intuitive three-tier labeling system of Safe, Caution, and Avoid provides users with immediately understandable and actionable health guidance. The detailed flagged ingredient explanations that accompany each label go beyond simply warning users about risks by educating them about the specific reasons behind each flag, building their nutritional knowledge and awareness over time with every scan they perform.

The AI-Powered Recommendation Module ensures that NutriSafe functions not merely as a system that tells users what they cannot eat but as a practical, constructive, and empowering health companion that actively helps them find better and safer alternatives. By automatically generating personalized lists of compatible food products whenever an incompatible item is detected, the module converts what could easily be a frustrating and discouraging experience into a helpful and confidence-building one. Users leave every incompatible scan interaction not just informed about a risk but equipped with practical options for addressing it.

The Symptom Feedback Module introduces a genuinely innovative and unique dimension to the NutriSafe system by creating a personal health feedback loop that no existing food inspection application currently offers. By collecting user-reported health reactions after food consumption and systematically linking them to specific ingredients, the module builds a continuously growing repository of personal health experience data that becomes progressively more valuable and insightful as the user continues to engage with the application over time.

The Health Risk Prediction Module transforms the accumulated symptom feedback data into proactive and personalized health intelligence through TensorFlow-based machine learning models. This module represents one of the most significant technical achievements of the NutriSafe project: the ability to warn users about potential adverse health reactions from new products they are considering consuming, based on patterns learned from their own historical symptom data. This proactive capability is the single feature that most fundamentally differentiates NutriSafe from all existing food inspection tools, every one of which is entirely reactive in nature and incapable of anticipating future risks.

The AI Chatbot Module ensures that NutriSafe is accessible and genuinely supportive for users across the full spectrum of technical and nutritional knowledge levels. By providing instant, conversational, plain-English dietary guidance through an NLP-powered assistant available at any point during the user's interaction with the application, the module ensures that no user is left without reliable dietary support regardless of how technically complex their question may be or how limited their background in nutritional science is.

The system was developed using the Agile Software Development Methodology combined with User-Centered Design principles, ensuring that the development process was both technically disciplined and continuously aligned with the real needs and expectations of the target user audience. The carefully chosen technology stack of Kotlin for the Android application, Python for all AI and machine learning components, Tesseract OCR for text extraction, and TensorFlow for deep learning model development proved to be a highly effective and well-integrated combination for delivering the complex, multi-disciplinary functionality that NutriSafe requires [8].

The comprehensive testing program conducted across unit, functional, integration, automated, performance, security, and usability evaluation levels confirmed that all functional and non-functional requirements documented in the Software Requirements Specification were successfully implemented and thoroughly verified in the final system. All 15 unit tests passed without exception, all 13 functional tests confirmed correct end-to-end behavior, and all 8 integration tests verified accurate cross-module data flow. Performance testing confirmed that the system meets its response time requirements for both scanning and chatbot interactions under normal operating conditions. Security testing confirmed that all sensitive user health data is properly encrypted at rest and in transit and that role-based access controls function correctly throughout the system. The usability evaluation confirmed that the interface is accessible, intuitive, and understandable for users of all technical backgrounds including those with no prior experience with nutritional or health technology applications.

Viewed in its entirety, NutriSafe successfully and convincingly demonstrates that artificial intelligence, machine learning, image processing, and natural language processing can be thoughtfully combined into a single, coherent, and genuinely impactful mobile application that addresses a real and important public health challenge. The system advances well beyond the capabilities of any existing food scanning application and establishes a new and higher standard for what a truly personalized, proactive, and intelligent food safety tool can and should provide to its users.

NutriSafe is not simply a food inspection application. It is a comprehensive, intelligent, proactive, and deeply personalized health companion that empowers individuals to take meaningful and informed control of their dietary choices and, through those choices, to support and protect their long-term health and wellbeing every single day.

7.2 Future Work

While NutriSafe delivers a complete, integrated, and fully functional system in its current version, the development team recognizes that there are several highly valuable enhancements and extensions that could further deepen the application's intelligence, broaden its accessibility, and increase its positive impact for a wider and more diverse range of users. The following future work directions have been identified through a combination of reflecting on the limitations of the current system, analyzing feedback collected during the testing and evaluation phase, and studying emerging trends in health technology, mobile AI, and personalized medicine.

7.2.1 Multi-Language Support

The current version of NutriSafe supports only the English language for both OCR text extraction from food packaging and NLP-based chatbot interaction. While this is appropriate for the initial release, it represents a significant limitation in terms of the application's global accessibility, particularly given that diet-related chronic illnesses are most prevalent in regions of the world where English is not the primary language.

A highly valuable and impactful future enhancement would be the addition of comprehensive multi-language support to make NutriSafe accessible to a much broader and more diverse global audience. Achieving this would require training and integrating multilingual OCR models capable of accurately recognizing and extracting ingredient text written in languages such as Urdu, Arabic, Hindi, French, Mandarin, and others from food packaging photographs. The NLP chatbot component would similarly need to be extended with multilingual language understanding models capable of processing and responding to user queries in multiple languages with the same accuracy and helpfulness as the current English-only implementation. The addition of multi-language support would represent a transformative step toward making NutriSafe a genuinely global and universally accessible food safety tool.

7.2.2 Wearable Device Integration

The current version of NutriSafe relies entirely on health data entered manually by the user through the health profile and symptom feedback features. While this approach is effective and provides a strong foundation for personalized analysis, it does not incorporate the real-time physiological data that modern wearable health monitoring devices can provide.

A powerful and technically exciting future enhancement would be the integration of NutriSafe with wearable health devices such as smartwatches and fitness trackers capable of providing real-time biometric readings including blood glucose levels, heart rate, blood pressure measurements, and physical activity data. By incorporating this continuous stream of real-time physiological information into the compatibility analysis and health risk prediction processes, NutriSafe could provide significantly more dynamic, responsive, and clinically relevant health assessments. For example, if a user's connected blood glucose monitor reports an elevated reading at the moment they are scanning a food product, the system could automatically apply a more conservative compatibility threshold for sugar and high-glycemic ingredients than it would under normal blood glucose conditions. This kind of real-time biometric-aware adaptive analysis would represent a major advancement in the personalization and clinical sophistication of the application.

7.2.3 Voice Input and Accessibility Enhancements

Adding voice input functionality to the NutriSafe application would make it substantially more accessible for several important and currently underserved user groups. Elderly users who find typing on small touchscreens challenging, individuals with motor impairments who have difficulty with precise touch input, and visually impaired users who rely on screen readers and audio feedback would all benefit enormously from the ability to interact with the chatbot, navigate between screens, and submit symptom reports using natural voice commands rather than text input.

Voice-activated food scanning would also be a valuable addition, allowing users to describe a food item or ingredient verbally and have the system retrieve and analyze the relevant data without requiring them to photograph a product label. In addition to voice input, future versions of NutriSafe should incorporate enhanced accessibility features including full screen reader compatibility, customizable high-contrast display modes for users with visual impairments, adjustable font sizes throughout the interface, and haptic feedback for important system alerts. These enhancements would ensure that NutriSafe can be used effectively and comfortably by every member of its intended audience without exception.

7.2.4 Comprehensive Daily Nutritional Tracking

The current version of NutriSafe focuses on evaluating the safety and compatibility of individual food products at the point of scanning. While this is highly valuable for preventing harmful consumption decisions, it does not provide the user with a broader picture of their overall dietary health across an entire day, week, or month.

A significant and highly useful future enhancement would be the development of a comprehensive daily nutritional tracking module that automatically aggregates the nutritional data from all food products the user has scanned and consumed throughout each day. This module would monitor the user's cumulative daily intake of calories, macronutrients including carbohydrates, proteins, and fats, and key micronutrients including sodium, potassium, vitamins, and minerals, and compare these cumulative values against the recommended daily intake thresholds appropriate for the user's specific health conditions. Users could receive proactive notifications when their cumulative intake of a particular nutrient approaches a threshold that is relevant to their health condition, for example when a user with hypertension is approaching their recommended daily sodium limit, even before they consume another product that might push them over it. This feature would transform NutriSafe from a point-of-purchase food safety checker into a continuous, holistic, and genuinely comprehensive daily dietary management companion.

7.2.5 Integration with Healthcare Professionals

The current version of NutriSafe operates as a standalone consumer application without any formal connection to the healthcare system or to the registered medical and nutritional professionals who manage the health conditions of many of its target users. This represents both a limitation and an opportunity for future development.

A valuable and clinically significant future enhancement would be the creation of a dedicated healthcare professional access module that allows registered dietitians, nutritionists, and medical doctors to securely access and review their patients' NutriSafe food scan logs, compatibility reports, symptom feedback history, health risk prediction results, and nutritional tracking data with the patient's explicit consent. This professional access layer would allow healthcare providers to monitor trends in their patients' food consumption and health reactions between clinical appointments, provide expert dietary advice and updated health guidance directly within the NutriSafe platform, review and where necessary adjust AI-generated recommendations based on clinical judgment, and set personalized health thresholds and alerts for individual patients based on their current clinical status. The integration of professionally validated medical oversight into the AI-driven system would significantly strengthen user trust in NutriSafe's recommendations and establish the application as a legitimate, clinically supported, and medically endorsed health management tool rather than simply a consumer technology product.

7.2.6 Expanded Machine Learning Model Training

The health risk prediction machine learning models in the current version of NutriSafe are trained individually for each user based on their own personal symptom feedback data. This approach produces highly personalized predictions but is constrained in accuracy during the early stages of a user's engagement with the application when the personal training dataset is still small.

A significant technical future enhancement would be the adoption of a federated learning approach to supplement individual user models with aggregate population-level knowledge derived from the collective, privacy-preserved symptom feedback data of all consenting NutriSafe users. In federated learning, model training occurs locally on each user's device using their personal data, and only the learned model parameters rather than the underlying personal data are shared with the central server. These aggregated parameters can then be used to improve the accuracy and generalizability of predictions for all users, including new users who have not yet accumulated enough personal feedback data to support reliable individual predictions. This approach would make NutriSafe's health risk prediction capability significantly more powerful and accurate from the very first scan while fully preserving the privacy and confidentiality of every individual user's personal health data.

7.2.7 Offline Functionality

The current version of NutriSafe requires an active internet connection for all of its core features including OCR processing, compatibility analysis, recommendation retrieval, and chatbot interaction. While internet connectivity is widely available in most urban environments, this dependence on network access limits the application's reliability and usefulness in locations with poor or no coverage, including rural areas, certain indoor environments, and regions with limited mobile data infrastructure.

A meaningful and practically important future enhancement would be the development of a lightweight offline mode for NutriSafe that allows users to perform basic food scanning and compatibility checking using a locally stored compressed subset of the ingredient safety database and a lightweight on-device machine learning model that does not require a server connection to function. While the offline mode would necessarily be more limited in scope than the full online version, it would ensure that users can still access the most critical food safety information, particularly allergy and serious contraindication warnings, even when they are not connected to the internet. Providing reliable offline functionality would significantly improve the application's real-world robustness and make it genuinely useful in a much wider range of everyday situations and environments.

7.3 Final Remarks

The NutriSafe project has been a comprehensive, technically challenging, and deeply meaningful academic and professional undertaking that has drawn upon and integrated knowledge from virtually every major discipline covered throughout the four-year Bachelor of Science in Computer Science program. It has demonstrated in concrete and functional terms how artificial intelligence, machine learning, image processing, natural language processing, mobile application development, database management, software engineering, and human-computer interaction can be thoughtfully combined and directed toward solving a genuinely important real-world challenge that affects the daily lives of millions of people.

The system that has been produced through this project is not a theoretical proof of concept or a limited demonstration prototype. It is a complete, rigorously tested, and fully functional application that integrates seven sophisticated AI-driven modules into a unified and accessible mobile experience. Every design decision, every algorithm, every database relationship, and every user interface element was made with a clear and specific purpose grounded in the documented requirements and the real needs of the target users.

The future work directions identified in this chapter represent a rich, exciting, and well-motivated roadmap for the continued evolution of NutriSafe beyond its current version. Each proposed enhancement builds meaningfully and logically on the strong foundation that the current system has established, whether by deepening the intelligence of existing features through expanded machine learning capabilities, broadening the accessibility of the application through multi-language and voice support, extending its clinical relevance through wearable integration and healthcare professional connectivity, or improving its reliability through offline functionality. Together, these future directions point toward a version of NutriSafe that could grow from a final year project into a genuinely impactful, widely adopted, and clinically recognized health technology product.

The development team is proud of what has been achieved through this project and is grateful for the guidance, support, and encouragement provided by the project supervisor, the Department of Computer Science, and all those who contributed to making NutriSafe a reality. The team is confident that NutriSafe, in its current form and in its future evolution, has the genuine potential to make a meaningful, lasting, and positive difference in the health and wellbeing of the people it serves.

Chapter 8
References

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